Metric Spaces & Topology: Sections 0–4, fully explained

Every definition, theorem and proof from the lecture notes, plus a complete worked solution to every exercise (the numbered ones and the ones in the text), written in the terminology an examiner expects to see.

How to use this guide (a 3–4 hour plan)

Time	What to do
0:00–0:15	Read the Proof toolkit and Section 0. Learn the axioms (MS1)–(MS3) by name.
0:15–1:00	Section 1. Try each exercise *before* opening its solution. Learn Cantor's theorem and the countability rules.
1:00–1:45	Section 2. Learn the LUB Principle, Approximation of suprema, the Sandwich Principle, BWT and the Uniform Continuity Theorem. Most later proofs use these.
1:45–2:30	Section 3. Practise checking the metric axioms, and learn Lipschitz maps, isometries, dist_A, balls and sequences.
2:30–3:30	Section 4, the heaviest part: open, closed, closure, interior, boundary, continuity via open sets, the two kinds of equivalence of metrics, and subspaces.
3:30–4:00	Trap questions and flashcards. Re-do any exercise you got wrong.

Colour key: ■ Definition · ■ Theorem/Proposition · ■ Exercise · ■ Pitfall · ■ Exam phrasing. Solutions are folded; click to open. The sidebar buttons open or close all of them at once.

Notation used throughout (as in the notes): $\mathbb{N} = \{1, 2, 3, \dots\}$, $\mathbb{Z}_+ = \{0, 1, 2, \dots\}$, $]a, b[$ is an open interval, $\mathcal{F}(S; T)$ is the set of all functions $S \to T$, $\mathcal{P}(S)$ is the power set, and "TFAE" means "the following are equivalent".

Proof toolkit & exam language

HOW EXAMINERS WANT PROOFS WRITTEN

- **Name the axiom or result you use**, e.g. "by (MS3), the Triangle Inequality", "by the Approximation of suprema", "by the Sandwich Principle", "by the LUB Principle", "by Bolzano–Weierstrass".
- **TFAE proofs**: prove a cycle (i)⇒(ii)⇒(iii)⇒(i). Say so at the start.
- **Set equality** $A = B$: prove $A \subset B$ *and* $B \subset A$ (an "element chase": "let $x \in A$ … hence $x \in B$").
- **Contrapositive**: to prove $P \Rightarrow Q$, prove $\neg Q \Rightarrow \neg P$. **Contradiction**: assume P and $\neg Q$ and derive something false.
- **Well-defined**: when you define a function on equivalence classes (or by a formula that might be ambiguous), check that the value does not depend on the choice of representative.
- **Quantifier order matters**: "$\forall \varepsilon > 0\ \exists \delta > 0$" (continuity) differs from "$\exists \delta > 0\ \forall x$" (uniform continuity: δ independent of the point).
- **ε-room**: to show $a \leq b$, it suffices to show $a < b + \varepsilon$ for every $\varepsilon > 0$. To show $a = 0$ for $a \geq 0$, show $a < \varepsilon$ for all $\varepsilon > 0$.
- **Sup/inf**: to show $s = \sup A$, show (1) s is an upper bound and (2) nothing smaller is: $\forall \varepsilon > 0\ \exists a \in A$ with $a > s - \varepsilon$.

THE BIG DEPENDENCY CHAIN (MEMORISE THE ORDER)

LUB Principle (axiom) ⇒ Archimedean property ⇒ density of $\mathbb{Q}$

LUB ⇒ bounded monotone ⇒ convergent + monotone subsequence ⇒

Bolzano–Weierstrass ⇒ Uniform Continuity Thm (UC, bounded, max/min) ⇒

IVT corollary

lim sup / lim inf ⇒ Completeness of $\mathbb{R}$ (Cauchy ⇔ convergent)

Metric axioms ⇒ balls, open sets ⇒ closed = complement of open ⇒

continuity ⇔ preimages of open sets are open ⇒ equivalence of metrics

Section 0: What is a metric space?

Foundation for everything that follows. You asked for Sections 1–4, but each of them uses these axioms, so this section is short and should be learnt first.

BIG PICTURE

Linear algebra abstracts "adding and scaling" into the axioms of a **vector space**. The structure-preserving maps are **linear maps** and **linear isomorphisms**. Distance geometry abstracts "distance between points" into three axioms, giving a **metric space**. Its maps are **Lipschitz maps** and **isometries**. Everything later (convergence, continuity, open and closed sets) comes from these three axioms.

Examples listed in the notes: the Euclidean, Manhattan (ℓ^1) and Chebyshev (ℓ^∞) metrics on $\mathbb{R}^N$; the Hausdorff distance between sets; distances between functions; the p-adic distance on $\mathbb{Q}$; the Hamming distance between strings; the graph distance; matrix distances (rank, Frobenius/Hilbert–Schmidt, operator norm); the geodesic distance on a manifold; the discrete metric.

DEFINITION (METRIC, METRIC SPACE)

A **metric** on a set S is a function $d : S \times S \to [0, \infty[$ satisfying

(MS1) $\forall x, y \in S :\ d(x, y) = 0 \iff x = y$;

(MS2) $\forall x, y \in S :\ d(y, x) = d(x, y)$ **(Symmetry)**;

(MS3) $\forall x, y, z \in S :\ d(x, z) \leq d(x, y) + d(y, z)$ **(Triangle Inequality)**.

A **metric space** E is a set S together with a specified metric d on it: $E = (S, d)$.

Refined version (useful for checking axioms one at a time): (MS0) $0 \leq d(x, y) < \infty$ **(Positivity & Finiteness)**; (MS1a) $d(x, x) = 0$; (MS1b) $d(x, y) = 0 \Rightarrow x = y$ **(Definiteness)**; plus (MS2) and (MS3).

PITFALL

The same set can carry different metrics, giving *different* metric spaces: $(\mathbb{R}, d_{\text{standard}})$ and $(\mathbb{R}, d_{\text{discrete}})$ have the same underlying set but very different properties.

Motivating examples

EXAMPLE (STANDARD METRIC ON $\mathbb{R}$)

$d(x,y) := |y-x|$. (MS1) holds because $|a| = 0 \iff a = 0$. (MS2) holds because $|-a| = |a|$. (MS3) holds because $|a+b| \le |a| + |b|$. All three come from $|a| := \max\{a, -a\}$ and the fact that $\mathbb{R}$ is an **ordered field**. ($\mathbb{R}$ is the unique **complete ordered field** up to isomorphism.)

EXAMPLE (EUCLIDEAN METRIC D_2 ON $\mathbb{R}^N$)

$d_2(x,y) = \sqrt{(y_1-x_1)^2 + \cdots + (y_N - x_N)^2}$. (MS3) follows from the **Cauchy–Schwarz Inequality** $\sum x_i y_i \le \sqrt{\sum x_i^2}\sqrt{\sum y_i^2}$. The **Euclidean norm** is $\|x\|_2 = \sqrt{\sum x_i^2}$, and $(\mathbb{R}^N, d_2)$ is written $\mathbb{E}^N$.

▾ Why Cauchy–Schwarz gives the Triangle Inequality

$\|a+b\|_2^2 = \|a\|_2^2 + 2\sum a_i b_i + \|b\|_2^2 \le \|a\|_2^2 + 2\|a\|_2\|b\|_2 + \|b\|_2^2 = (\|a\|_2 + \|b\|_2)^2$. Take square roots and put $a = y-x$, $b = z-y$: $d_2(x,z) = \|a+b\|_2 \le d_2(x,y) + d_2(y,z)$.

EXAMPLE (DISCRETE METRIC)

$d(s,t) = 1$ if $s \ne t$, and 0 if $s = t$. It works on *any* set S (Exercise 0.1).

EXAMPLE (COUNTING METRIC ON $\mathcal{P}(S)$, S FINITE)

$d(A,B) := \#(A \Delta B)$, where $A\Delta B = (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$ is the **symmetric difference**. (MS1): $\#(A\Delta B) = 0 \iff A\Delta B = \emptyset \iff A \cup B = A \cap B \iff A = B$. (MS2): $B\Delta A = A\Delta B$. If S is infinite, the same formula is a metric on $\mathcal{P}_{00}(S) = \{A \subset S : \#A < \infty\}$, the **finite power set**.

EXAMPLE (HAUSDORFF METRIC)

$\mathcal{K}(\mathbb{R}^2)$ is the collection of subsets of $\mathbb{R}^2$ that are (1) nonempty, (2) bounded, and (3) closed. The **δ-fattening** of K is $K^\delta := \{x \in \mathbb{R}^2 : \exists a \in K \; d_2(x,a) \le \delta\}$, and

$$d_{\text{Hausdorff}}(K,J) := \inf\{\delta \ge 0 : K^\delta \supset J \text{ and } J^\delta \supset K\}.$$

Why each assumption is needed: **boundedness** (2) makes the set of admissible δ nonempty, so $d < \infty$ (MS0). **Closedness** (3) is needed for (MS1b): if $x \in K \setminus J$ then $\operatorname{dist}(x,J) > 0$ because J is closed, so for $\delta = \operatorname{dist}(x,J)/2$ we have $J^\delta \not\supset K$, which forces $d(J,K) > 0$. (MS1a) holds because $K^\delta \supset K$ for all δ. (MS2) is clear because the definition is symmetric in K and J. The same construction works in $\mathbb{E}^N$.

EXAMPLE (INTEGRATION METRIC D_1 ON $C(I;\mathbb{R})$, $I=[0,1]$)

$d_1(f,g)=\int_0^1|g-f|$. (MS0): $|g-f|$ is continuous, hence Riemann integrable, with a nonnegative finite integral. (MS1a): $\int 0=0$. (MS2): $|-h|=|h|$. (MS3): $|k+l|\le|k|+|l|$ pointwise, so $\int|k+l|\le\int|k|+\int|l|$ by order preservation and linearity. (MS1b) **needs continuity** (Exercise 0.2).

EXAMPLE (UNIFORM / SUP METRIC ON $C(I;\mathbb{R})$)

$d_{\sup}(f,g)=\sup\{|g(s)-f(s)|:s\in I\}$. It is finite because a continuous function on a closed bounded interval is bounded. (MS1a) and (MS2) are easy. (MS3) and (MS1b) are Exercise 0.3.

EXAMPLE (TWO NONSTANDARD METRICS ON $\mathbb{R}^2$, "TGV" METRICS)

$d_{TGV}(u,v)=\|u\|_2+\|v\|_2$ if $u\neq v$, and 0 if $u=v$. Picture: every journey goes via the hub 0 ("via Paris").

$d'_{TGV}(u,v)=\|u\|_2+\|v\|_2$ if u,v are linearly independent, and $\|v-u\|_2$ otherwise. Picture: you may travel straight along a line through 0, otherwise you go via 0.

(MS1): $\|u\|+\|v\|=0\Rightarrow u=v=0$. (MS2): $\|-x\|=\|x\|$. The Triangle Inequality is Exercise 0.5.

Exercises for Section 0, with full solutions

EXERCISE 0.1 (DISCRETE METRIC)

10 marks

Prove the discrete metric on a set S is a metric.

▾ Solution

d takes values in $\{0,1\}\subset[0,\infty[$ (MS0).

(MS1) By definition, $d(s,t)=0$ exactly when $s=t$.

(MS2) $s\neq t\iff t\neq s$, so $d(t,s)=d(s,t)$.

(MS3) Let $x,y,z\in S$. If $x=z$ then $d(x,z)=0\le d(x,y)+d(y,z)$. If $x\neq z$ then y cannot equal both x and z, so $y\neq x$ or $y\neq z$. Hence $d(x,y)+d(y,z)\ge 1=d(x,z)$. ∎

EXERCISE 0.2 ((MS1B) FOR D_1)

10 marks

Show: $h \in C(I;\mathbb{R})$ and $\int |h| = 0 \Rightarrow h = 0$.

▾ Solution

We prove the **contrapositive**: $h \neq 0 \Rightarrow \int |h| > 0$. Suppose $h(p) \neq 0$ for some $p \in I$, and let $c := |h(p)| > 0$. The function $|h|$ is continuous at p, so with $\varepsilon = c/2$ there is $\delta > 0$ such that $s \in I,\ |s - p| < \delta \Rightarrow \big||h(s)| - c\big| < c/2$, so $|h(s)| > c/2$.

Let $J = [p - \delta/2, p + \delta/2] \cap I$. This is a closed subinterval of I of length $\ell \geq \min\{\delta/2, 1\} > 0$, and $|h| \geq c/2$ on J. Since $|h| \geq 0$ everywhere, order preservation and additivity of the integral give $\int_I |h| \geq \int_J |h| \geq \frac{c}{2}\,\ell > 0$. ∎

Continuity is essential: $h = \mathbf{1}_{\{0\}}$ has $\int |h| = 0$ but $h \neq 0$ (that h is Riemann integrable but not continuous).

EXERCISE 0.3 ((MS3) AND (MS1B) FOR D_{SUP})

10 marks

▾ Solution

(MS3) Let $f, g, h \in C(I;\mathbb{R})$ and $s \in I$. By the Triangle Inequality in $\mathbb{R}$, $|h(s) - f(s)| \leq |g(s) - f(s)| + |h(s) - g(s)| \leq d_{\sup}(f,g) + d_{\sup}(g,h)$. The right-hand side does not depend on s, so it is an upper bound for $\{|h(s) - f(s)| : s \in I\}$. Taking the supremum gives $d_{\sup}(f,h) \leq d_{\sup}(f,g) + d_{\sup}(g,h)$.

(MS1b) If $d_{\sup}(f,g) = 0$ then for every s, $0 \leq |g(s) - f(s)| \leq 0$, so $f(s) = g(s)$. Hence $f = g$. (Continuity is not needed here.) ∎

EXERCISE 0.4 (TRIANGLE INEQUALITY FOR THE COUNTING METRIC)

10 marks

▾ Solution

Claim: $A \Delta C \subset (A \Delta B) \cup (B \Delta C)$. Let $x \in A \Delta C$. By symmetry of roles, assume $x \in A \setminus C$. If $x \in B$ then $x \in B \setminus C \subset B \Delta C$. If $x \notin B$ then $x \in A \setminus B \subset A \Delta B$. The case $x \in C \setminus A$ is the same with letters swapped.

So, for finite sets, $\#(A \Delta C) \leq \#\big((A \Delta B) \cup (B \Delta C)\big) \leq \#(A \Delta B) + \#(B \Delta C)$, i.e. $d(A,C) \leq d(A,B) + d(B,C)$. ∎

EXERCISE 0.5 (TRIANGLE INEQUALITY FOR D_{TGV} AND D'_{TGV})

20 marks

▾ Solution

Write $\|\cdot\|$ for $\|\cdot\|_2$. Let $u, v, w \in \mathbb{R}^2$.

d_{TGV}. If $u = w$ the left side is 0. Suppose $u \neq w$, so $d(u,w) = \|u\| + \|w\|$. If $v = u$, the right side is $0 + d(u,w)$, which gives equality. If $v = w$, likewise. Otherwise $v \neq u$ and $v \neq w$, and the right side is $\|u\| + 2\|v\| + \|w\| \geq \|u\| + \|w\|$.

d'_{TGV}. First a useful fact: in every case $\|v - u\| \leq d'(u,v)$, because $\|v-u\| \leq \|u\| + \|v\|$.

Case A: u, w linearly dependent. Then $d'(u,w) = \|w-u\| \leq \|v-u\| + \|w-v\| \leq d'(u,v) + d'(v,w)$.

Case B: u, w linearly independent. Then $u \neq 0 \neq w$ and $d'(u,w) = \|u\| + \|w\|$.

- If $v = 0$: the pairs $\{u, 0\}$ and $\{0, w\}$ are dependent, so the right side is $\|u\| + \|w\|$, which gives equality.
- If $v \neq 0$, then u, v dependent and v, w dependent together would put u, v, w on one line through 0, contradicting independence of u, w. So at most one of the pairs is dependent.
  - Neither dependent: the right side is $\|u\| + 2\|v\| + \|w\| \geq \|u\| + \|w\|$.
  - u, v dependent, v, w independent: the right side is $\|v-u\| + \|v\| + \|w\| \geq \|u\| + \|w\|$, since $\|u\| \leq \|u - v\| + \|v\|$.
  - The remaining case is symmetric.

∎

EXERCISE 0.6 ((MS3) FOR THE HAUSDORFF METRIC)

10 marks

(a) For $A, B \subset \mathbb{R}^N$ and $\gamma, \delta > 0$: (i) $A \subset B \Rightarrow A^\delta \subset B^\delta$; (ii) $(A^\gamma)^\delta \subset A^{\gamma+\delta}$. (b) Deduce (MS3).

▼ Solution

(a)(i) If $x \in A^\delta$, there is $a \in A$ with $d_2(x,a) \le \delta$. Since $a \in B$, $x \in B^\delta$.

(a)(ii) If $x \in (A^\gamma)^\delta$, there is $y \in A^\gamma$ with $d_2(x,y) \le \delta$, and there is $a \in A$ with $d_2(y,a) \le \gamma$. By (MS3), $d_2(x,a) \le \gamma + \delta$, so $x \in A^{\gamma+\delta}$.

(b) Let $K, J, L \in \mathcal{K}(\mathbb{R}^N)$ and $\varepsilon > 0$. Write $\Delta(K,J) = \{\delta : K^\delta \supset J,\ J^\delta \supset K\}$. By (a)(i) this set is "upward closed": if $\delta \in \Delta$ and $\delta' > \delta$ then $\delta' \in \Delta$. By the approximation property of infima, pick $\gamma \in \Delta(K,J)$ with $\gamma < d(K,J) + \varepsilon$ and $\delta \in \Delta(J,L)$ with $\delta < d(J,L) + \varepsilon$ (both can be taken > 0). Then:

$L \subset J^\delta$ and $J \subset K^\gamma \Rightarrow J^\delta \subset (K^\gamma)^\delta \subset K^{\gamma+\delta}$, so $L \subset K^{\gamma+\delta}$.

$K \subset J^\gamma$ and $J \subset L^\delta \Rightarrow J^\gamma \subset (L^\delta)^\gamma \subset L^{\gamma+\delta}$, so $K \subset L^{\gamma+\delta}$.

Hence $\gamma + \delta \in \Delta(K,L)$, so $d(K,L) \le \gamma + \delta < d(K,J) + d(J,L) + 2\varepsilon$. Since $\varepsilon > 0$ was arbitrary, $d(K,L) \le d(K,J) + d(J,L)$. ∎

Section 1: Sets, functions & countability

The language of the whole course. Expect exam questions on images and inverse images, injectivity and surjectivity, equivalence relations, countability and Cantor's theorem.

1.1 Sets and the Axiom of Choice

DEFINITIONS

- **Empty set** $\emptyset$ (a letter from the Danish alphabet, not the Greek ϕ).
- $A \subset S$ (**subset**): $x \in A \Rightarrow x \in S$. $A \subsetneq S$ (**proper subset**): $A \subset S$ and $A \neq S$.
- **Set difference** $A \setminus B := \{s \in A : s \notin B\} = A \cap (S \setminus B)$. Here $S \setminus B$ is the **complement** of B in the universe S, also written $B^{\complement}$.
- **Power set** $\mathcal{P}(S) = \{A : A \subset S\}$.
- For a family $\mathcal{C}$ of subsets of S: $\bigcup \mathcal{C} := \{s \in S : \exists C \in \mathcal{C}\ s \in C\}$ and $\bigcap \mathcal{C} := \{s \in S : \forall C \in \mathcal{C}\ s \in C\}$.

DISTRIBUTIVE LAWS & DE MORGAN'S LAWS

$$A \cap \bigcup \mathcal{C} = \bigcup_{C \in \mathcal{C}} (A \cap C), \qquad A \cup \bigcap \mathcal{C} = \bigcap_{C \in \mathcal{C}} (A \cup C),$$

$$A \setminus \bigcup \mathcal{C} = \bigcap_{C \in \mathcal{C}} (A \setminus C), \qquad A \setminus \bigcap \mathcal{C} = \bigcup_{C \in \mathcal{C}} (A \setminus C).$$

Memory aid: taking a complement swaps $\cup$ and $\cap$.

TRICKY CASE: $\mathcal{C} = \emptyset$

$\bigcup \emptyset = \emptyset$ (no set contributes any element), but by convention $\bigcap \emptyset = S$ (every s vacuously belongs to "all" members). This is why the **universe** must be known.

CARTESIAN PRODUCTS

$S \times T = \{(s,t) : s \in S, t \in T\}$, and $S \times T = \emptyset \iff S = \emptyset$ or $T = \emptyset$. $S_1 \times \cdots \times S_N$ consists of N-tuples, and S^N is the case where all factors are S.

For an **indexed family** $(S_\lambda)_{\lambda\in\Lambda}$: $\prod_{\lambda\in\Lambda} S_\lambda$ is the set of all families $(s_\lambda)_{\lambda\in\Lambda}$ with $s_\lambda \in S_\lambda$ (generalised sequences). Equality is componentwise: $a = b \iff \forall\lambda\ a_\lambda = b_\lambda$. If all $S_\lambda = S$, the product is written S^Λ. So $\mathbb{R}^\mathbb{N}$ is the set of all real sequences.

Strange cases. If $\Lambda = \emptyset$, the product has exactly **one** element (the empty family). If some $S_\mu = \emptyset$, the product is **empty** ("s_μ has nowhere to go").

AXIOM OF CHOICE (AC)

If $(S_\lambda)_{\lambda\in\Lambda}$ is a family of nonempty sets, then $\prod_\lambda S_\lambda \neq \emptyset$.

Equivalent form (choice function): for every indexed family $\{C_\lambda\}$ of nonempty subsets of S there is $f \in \mathcal{F}(\Lambda; S)$ with $f(\lambda) \in C_\lambda$ for all λ.

AC is **not deducible** from the other axioms of set theory, but it (or an equivalent) is used throughout modern mathematics.

1.2 Functions, AC again & images

FUNCTION

A function f has **three ingredients**: a **domain** (source) S, a **codomain** (target) T, and an unambiguous rule assigning to each $s \in S$ a unique $f(s) \in T$. We write $f : S \to T,\ s \mapsto f(s)$, and $\mathcal{F}(S;T)$ for the set of all such functions. Two functions are equal when all three ingredients agree.

Examples: $U \mapsto \dim U$ on the subspaces of V; the zero-set map $f \mapsto \{s : f(s) = 0\}$; $\det : M_n(\mathbb{R}) \to \mathbb{R}$.

BASIC CLASSES OF FUNCTION

1. **Identity** $\mathrm{id}_S : s \mapsto s$.
2. **Inclusion** $\iota_{A\to S} : A \to S,\ a \mapsto a$.
3. **Indicator function** $\mathbf{1}_A : S \to \mathbb{R}$, equal to 1 on A and 0 off A.
4. **Constant function** equal to t_0.
5. **Canonical projection** $\prod_j S_j \to S_k,\ (s_j) \mapsto s_k$.

Strange cases: $\mathcal{F}(\emptyset;T)$ has exactly one element (the empty function, whose rule is vacuous). If $S \neq \emptyset$ then $\mathcal{F}(S;\emptyset) = \emptyset$. If $\#S = n$ and $\#T = m$ then $\#\mathcal{F}(S;T) = m^n$, consistent with $m^0 = 1$ and $0^n = 0$ for $n \neq 0$.

RANGE, COMPOSITION, RESTRICTION, EXTENSION, CO-RESTRICTION

- **Range** (image of S under f): $\operatorname{Ran} f = \{f(s) : s \in S\} = \{t \in T : \exists s\ f(s) = t\}$. For example $\operatorname{Ran} \mathbf{1}_A$ is $\{1\}$ if $A = S$, $\{0\}$ if $A = \emptyset$, and $\{0, 1\}$ otherwise. f is constant t_0 iff $\operatorname{Ran} f = \{t_0\}$.
- **Composition** $(g \circ f)(s) = g(f(s))$. It is **associative**, but usually $f \circ g \neq g \circ f$ (e.g. $\sin(x^2) \neq (\sin x)^2$). Also $f \circ \mathrm{id}_S = f = \mathrm{id}_T \circ f$.
- **Restriction** $f|_A = f \circ \iota_{A \to S}$. An **extension** of $g \in \mathcal{F}(A; T)$ is any f with $f|_A = g$.
- **Co-restriction**: if $B \supset \operatorname{Ran} f$, then $f|^B : S \to B$, $s \mapsto f(s)$, satisfies $\iota_{B \to T} \circ f|^B = f$.

INJECTIVE, SURJECTIVE, BIJECTIVE

- **Injective**: $f(s_1) = f(s_2) \Rightarrow s_1 = s_2$. Equivalently (by the contrapositive) $s_1 \neq s_2 \Rightarrow f(s_1) \neq f(s_2)$.
- **Surjective**: $\operatorname{Ran} f = T$, i.e. $\forall t \in T\ \exists s \in S\ f(s) = t$.
- **Bijective**: both.

The codomain matters. $x \mapsto x^2$ is injective but not surjective as a map $\mathbb{R}_+ \to \mathbb{R}$; surjective but not injective as $\mathbb{R} \to \mathbb{R}_+$; bijective as $\mathbb{R}_+ \to \mathbb{R}_+$. Inclusions are injective. Canonical projections are surjective (if all factors are nonempty). Quotient maps are surjective.

Example. On $\mathbb{N}$, $f(n) = n + 1$ is injective and $g(n) = n - 1$ for $n \geq 2$, $g(1) = 1$ is surjective. Then $g \circ f = \mathrm{id}_{\mathbb{N}}$ is bijective, but $f \circ g$ is neither injective (since $(f \circ g)(1) = (f \circ g)(2) = 2$) nor surjective (it never takes the value 1).

PROPOSITION (UNIQUENESS OF INVERSES)

If $g \circ f = \mathrm{id}_S$ and $f \circ h = \mathrm{id}_T$, then $g = h$.

Proof. $g = g \circ \mathrm{id}_T = g \circ (f \circ h) = (g \circ f) \circ h = \mathrm{id}_S \circ h = h$, by associativity. ∎

PROPOSITION (INJECTIVITY/SURJECTIVITY & LEFT/RIGHT INVERTIBILITY)

Let $f \in \mathcal{F}(S;T)$.

(a)* f is injective $\iff$ there is a **left inverse** g ($g \circ f = \mathrm{id}_S$). [*Exclude the case $S = \emptyset \neq T$.]

(b) f is surjective $\iff$ there is a **right inverse** h ($f \circ h = \mathrm{id}_T$).

(c) f is bijective $\iff$ there is an **inverse** k ($k \circ f = \mathrm{id}_S$ and $f \circ k = \mathrm{id}_T$), written f^{-1}.

Proof.

The "if" parts follow from Exercise 1.4. For the "only if" parts (with $S, T \neq \emptyset$):

(a) Fix $s_0 \in S$. Define $g(t) = s$ if $t = f(s)$, and $g(t) = s_0$ if $t \notin \mathrm{Ran}\, f$. This is **well-defined** because injectivity makes the s with $f(s) = t$ unique. Then $g(f(s)) = s$.

(b) Each fibre $f^{-1}(\{t\})$ is nonempty. By the **Axiom of Choice** there is $h : T \to S$ with $h(t) \in f^{-1}(\{t\})$, so $f(h(t)) = t$.

(c) Combine (a), (b) and uniqueness of inverses. ∎

Remarks: left and right inverses are in general **not unique**. Even this basic result needs AC, in part (b).

COROLLARY (INJECTIONS ENTAIL SURJECTIONS AND VICE VERSA)

Avoiding the case $S = \emptyset \neq T$: there is an injection $S \to T$ $\iff$ there is a surjection $T \to S$. (A left inverse of an injection is a surjection, and a right inverse of a surjection is an injection.)

DIRECT AND INVERSE IMAGES

Direct image of $A \subset S$: $f(A) = \{f(s) : s \in A\}$, sometimes written $f[A]$. It defines a set map $\mathcal{P}(S) \to \mathcal{P}(T)$. $f(S) = \mathrm{Ran}\, f$.

Inverse image (preimage) of $C \subset T$: $f^{-1}(C) := \{s \in S : f(s) \in C\}$, a set map $\mathcal{P}(T) \to \mathcal{P}(S)$.

PITFALL: THE NOTATION $F^{-1}(C)$ DOES NOT MEAN F IS INVERTIBLE

Preimages exist for *every* function. Examples: $f^{-1}(\emptyset) = \emptyset$ and $f^{-1}(T) = S$. For $x \mapsto x^2$, $f^{-1}([1,4]) = [-2,-1] \cup [1,2]$. For f constant t_0, $f^{-1}(C) = S$ if $t_0 \in C$ and $\emptyset$ otherwise. If f is bijective with inverse g, then $f^{-1}(C) = g(C)$, so the two notations agree in that case.

IMAGES VS INVERSE IMAGES: THE RULES TO KNOW

Direct images (Exercise 1.8)	Inverse images (Prop. + Exercise 1.9)
$f(\bigcup \mathcal{A}) = \bigcup f(A)$ **(equality)**	$f^{-1}(\bigcup \mathcal{C}) = \bigcup f^{-1}(C)$
$f(\bigcap \mathcal{A}) \subset \bigcap f(A)$ **(only $\subset$)**	$f^{-1}(\bigcap \mathcal{C}) = \bigcap f^{-1}(C)$
$f(A \setminus B) \supset f(A) \setminus f(B)$ **(only $\supset$)**	$f^{-1}(D \setminus C) = f^{-1}(D) \setminus f^{-1}(C)$; $f^{-1}(T \setminus C) = S \setminus f^{-1}(C)$

Slogan: inverse images preserve every set operation. Direct images preserve only unions. This is why continuity is defined through *inverse* images of open sets in Section 4.

Counterexample showing $f(A \cap B) \subsetneq f(A) \cap f(B)$: $S = \{0, 1\}$, f constant, $A = \{0\}$, $B = \{1\}$. Then $f(A \cap B) = f(\emptyset) = \emptyset$, but $f(A) \cap f(B) = \operatorname{Ran} f \neq \emptyset$.

Proof. **(Inverse image of intersection.)** $s \in f^{-1}(\bigcap \mathcal{C}) \iff f(s) \in \bigcap \mathcal{C} \iff \forall C\ f(s) \in C \iff \forall C\ s \in f^{-1}(C) \iff s \in \bigcap f^{-1}(C)$. ∎

This uses the fact that $A = B$ iff $(\forall s\colon s \in A \iff s \in B)$.

Schröder–Bernstein theorem

THEOREM (SCHRÖDER–BERNSTEIN): YOU MUST KNOW THE STATEMENT

If there is an injection $S \to T$ and an injection $T \to S$, then there is a **bijection** $S \to T$.

The same conclusion holds with surjections both ways, or with an injection $S \to T$ together with a surjection $S \to T$.

The proof is **not examinable**. Idea: find $P \subset S$ with $h(T \setminus g(P)) = S \setminus P$. Then map P by g and $S \setminus P$ by h^{-1}. Such a P is a fixed point of the monotone map $\psi(A) = S \setminus h(T \setminus g(A))$ on $\mathcal{P}(S)$. The **Set Monotonicity Lemma** provides it: $P = \bigcap\{\psi(A) : \psi(A) \subset A\}$ satisfies $\psi(P) = P$.

1.3 Equivalence relations & partitions

DEFINITION

A relation $\sim$ on a nonempty set S is an **equivalence relation** if it is **reflexive** ($s \sim s$), **symmetric** ($s \sim t \Rightarrow t \sim s$) and **transitive** ($r \sim s \;\&\; s \sim t \Rightarrow r \sim t$).

The **equivalence class** is $[a]_\sim = \{s : s \sim a\}$. The **quotient set** is $S/\!\sim \;= \{[a] : a \in S\}$. The **quotient map** is $q : a \mapsto [a]$, and it is surjective. Key fact: $[a] = [b] \iff a \sim b$.

Examples: **equality** ($[a] = \{a\}$); the **full relation** ($[a] = S$); congruence mod p on $\mathbb{Z}$ ($n \equiv_p m$ iff $p \mid (m-n)$, with classes $p\mathbb{Z} + k$ and $\mathbb{Z} = \bigcup_{k=0}^{p-1} [k]_p$). The **empty relation** on a nonempty set is symmetric and transitive but *not reflexive*, so it is not an equivalence relation.

PARTITION

$\mathcal{C} \subset \mathcal{P}(S)$ is a **partition** of S if (1) $\bigcup \mathcal{C} = S$, (2) every $C \in \mathcal{C}$ is nonempty, and (3) $C \neq D \Rightarrow C \cap D = \emptyset$ (distinct blocks are disjoint).

THEOREM (PARTITIONS ≡ EQUIVALENCE RELATIONS)

(i) If $\sim$ is an equivalence relation, then $S/\!\sim$ is a partition. (ii) If $\mathcal{C}$ is a partition, then $s \sim_{\mathcal{C}} t :\iff \exists C \in \mathcal{C} :\; s, t \in C$ is an equivalence relation. (iii) These two procedures are mutually inverse.

Proof.

(i) By reflexivity $s \in [s]$, so the classes are nonempty and cover S. Suppose $r \in [s] \cap [t]$ and $u \in [t]$. Then $r \sim s$, $r \sim t$, $u \sim t$. By symmetry $u \sim t$, $t \sim r$, $r \sim s$, so by transitivity $u \sim s$. Hence $[t] \subset [s]$. Swapping the roles of s and t (using $r \in [t] \cap [s]$) gives $[s] \subset [t]$. So distinct classes are disjoint.

(ii) **Reflexive**: each s lies in some C, because $\bigcup \mathcal{C} = S$. **Symmetric**: obvious. **Transitive**: if $r, s \in C$ and $s, t \in D$, then $s \in C \cap D \neq \emptyset$, so $C = D$ and hence $r, t \in C$. ∎

THEOREM (CANONICAL DECOMPOSITION OF A FUNCTION)

For $f \in \mathcal{F}(S;T)$: (a) $s_1 \sim_f s_2 \iff f(s_1) = f(s_2)$ is an equivalence relation. (b) $\tilde{f} : S/\!\sim_f \to I := \operatorname{Ran} f$, $[s] \mapsto f(s)$, is a **well-defined bijection**. (c) $f = i \circ \tilde{f} \circ q$, where q is the quotient map and i is the inclusion.

Every function factorises as a surjection, then a bijection, then an injection. This is the basis of every "First Isomorphism Theorem" in algebra.

Proof. Well-defined: $[s_1] = [s_2] \Rightarrow s_1 \sim s_2 \Rightarrow f(s_1) = f(s_2)$. Injective: $f(s_1) = f(s_2) \Rightarrow s_1 \sim s_2 \Rightarrow [s_1] = [s_2]$. Surjective: $\operatorname{Ran} \tilde{f} = \operatorname{Ran} f = I$. Part (c) holds by definition. ∎

1.4 Countability and Cantor's theorem

FINITE, COUNTABLE

S is **finite** if $S = \emptyset$ or there is a bijection $S \to \{1, \dots, K\}$. The number K (0 for $\emptyset$) is unique and is called the **cardinality** $\#S$.

S is **countable** if there is an **injection** $f : S \to \mathbb{N}$.

RULES FOR COUNTABILITY (MEMORISE)

1. $\emptyset$ and $\mathbb{N}$ are countable.
2. For $S \neq \emptyset$: S is countable $\iff$ there is a **surjection** $\mathbb{N} \to S$.
3. Subsets of countable sets are countable.
4. If T is countable and there is an injection $S \to T$, or a surjection $T \to S$, then S is countable.
5. S is countable $\iff$ S is finite or there is a bijection $\mathbb{N} \to S$ (i.e. "countably infinite").
6. $\mathbb{N}^N$ is countable, via the injection $(k_1, \dots, k_N) \mapsto p_1^{k_1} \cdots p_N^{k_N}$ (the first N primes; uniqueness of prime factorisation).
7. **Finite products** of countable sets are countable: if $\phi_i : \mathbb{N} \to S_i$ are surjections, then $(n_i) \mapsto (\phi_i(n_i))$ is a surjection $\mathbb{N}^N \to \prod S_i$.
8. $\mathbb{Q}$ is countable: $\mathbb{Z} \times \mathbb{N} \to \mathbb{Q}$, $(n, m) \mapsto n/m$, is a surjection.
9. **A countable union of countable sets is countable**: with surjections $\psi_n : \mathbb{N} \to C_n$, the map $(n, m) \mapsto \psi_n(m)$ is a surjection $\mathbb{N} \times \mathbb{N} \to \bigcup C_n$. (Choosing all the ψ_n at once uses AC.)

CANTOR'S THEOREM (TWO GUISES)

(1) There is **no surjection** $S \to \mathcal{P}(S)$, and hence no injection $\mathcal{P}(S) \to S$.

Proof. Suppose $\phi : S \to \mathcal{P}(S)$ is surjective. Let $D := \{s \in S : s \notin \phi(s)\}$, the **diagonal set**. Then $D = \phi(x)$ for some x. If $x \in D$ then $x \notin \phi(x) = D$, a contradiction. If $x \notin D$ then $x \in \phi(x) = D$, a contradiction. ∎ (This is reminiscent of **Russell's Paradox**.)

(2) There is no surjection $S \to \mathcal{F}(S; \{0, 1\})$, and no injection the other way.

Proof. Given ϕ, write $\phi_s = \phi(s)$ and define the **diagonal function** $g(s) = 1 - \phi_s(s)$. Then $g(s) \neq \phi_s(s)$ for every s, so $g \neq \phi_s$ for all s, and $g \notin \operatorname{Ran} \phi$. ∎

Consequences. Using the bijections $\mathcal{F}(S; T) \cong T^S$ and $\mathcal{P}(S) \cong \mathcal{F}(S; \{0, 1\})$ (Exercises 1.6–1.7), the sets $\mathcal{P}(\mathbb{N}) \cong \mathcal{F}(\mathbb{N}; \{0, 1\}) \cong \{0, 1\}^{\mathbb{N}}$ are **uncountable**. Hence $\mathbb{R}$ is uncountable (Exercises 1.18 and 4.12). This was the key to Cantor's theory of infinite cardinals.

THEOREM (COUNTABLY INFINITE SETS ARE BIJECTIVE TO $\mathbb{N}$)

If S is countable and infinite, then there is a bijection $S \to \mathbb{N}$.

Proof. Let $g : S \to \mathbb{N}$ be an injection and $I = g(S)$, which is infinite. Using the **well-ordering of $\mathbb{N}$**, define $i_1 = \min I$ and $i_n = \min(I \setminus \{i_1, \ldots, i_{n-1}\})$. Then $i_1 < i_2 < \cdots$ and $I = \{i_k\}$, so $k \mapsto i_k$ is a bijection $\mathbb{N} \to I$. Compose its inverse with the co-restriction $S \to I$ of g. ∎

Corollary (Schröder–Bernstein for $\mathbb{N}$). An injection $S \to \mathbb{N}$ together with a surjection $S \to \mathbb{N}$ gives a bijection $S \to \mathbb{N}$. (S is countable, and it is infinite because it maps onto $\mathbb{N}$.)

1.5 Partial orders & Zorn's Lemma (self-study, non-examinable)

DEFINITIONS

Partial order $\leq$: reflexive, **antisymmetric** ($s \leq t$ & $t \leq s \Rightarrow s = t$), transitive. An **upper bound** u of A satisfies $a \leq u$ for all $a \in A$. A **maximal element** x satisfies $s \geq x \Rightarrow s = x$. A **chain** is a totally ordered subset. A **least upper bound** is an upper bound below every other upper bound; it is unique by antisymmetry.

Strict partial order $<$: $s < t \Rightarrow t \not< s$, plus transitivity. Moving between $\leq$ and $<$ ($s < t$ iff $s \leq t$ and $s \neq t$) is a pair of mutually inverse procedures.

Total order: a partial order with **comparability** ($s \leq t$ or $t \leq s$). **Strict total order**: **trichotomy** (exactly one of $s < t$, $s = t$, $s > t$) plus transitivity.

Zorn's Lemma: if every chain in a poset has an upper bound, there is a maximal element. It is equivalent to AC. Application: the **Hamel Basis Theorem** (every vector space has a basis): apply Zorn to the linearly independent subsets ordered by inclusion, where a maximal element is a basis.

Exercises for Section 1, with full solutions

EXERCISE 1.1 (SIMPLE INTERSECTION/UNION PROBLEM)

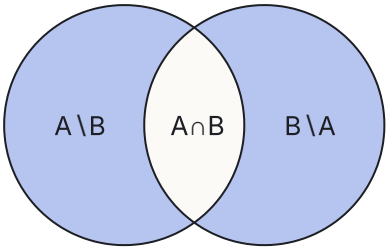
(a) TFAE: (i) $A \subset B$; (ii) $A \cap B = A$; (iii) $A \cup B = B$. (b) $(A \cup B) \setminus (A \cap B) = (A \setminus B) \cup (B \setminus A) =: A \Delta B$. (c) Draw it.

▾ Solution

(a) **(i)⇒(ii)**: $A \cap B \subset A$ always. If $x \in A$ then $x \in B$ by (i), so $x \in A \cap B$. **(ii)⇒(iii)**: $B \subset A \cup B$ always. If $x \in A \cup B$, either $x \in B$, or $x \in A = A \cap B \subset B$. **(iii)⇒(i)**: $x \in A \Rightarrow x \in A \cup B = B$.

(b) $x \in (A \cup B) \setminus (A \cap B) \iff x$ is in at least one of A, B but not both $\iff x$ is in exactly one of them $\iff$ ($x \in A$ and $x \notin B$) or ($x \in B$ and $x \notin A$) $\iff x \in (A \setminus B) \cup (B \setminus A)$.

(c) The shaded region is $A \Delta B$:



EXERCISE 1.2 (DISTRIBUTIVITY AND DE MORGAN)

Verify (a) $A \cap \bigcup \mathcal{C} = \bigcup_C (A \cap C)$ and (b) $A \setminus \bigcap \mathcal{C} = \bigcup_C (A \setminus C)$.

▾ Solution

(a) $x \in A \cap \bigcup \mathcal{C} \iff x \in A$ and $\exists C \in \mathcal{C} : x \in C \iff \exists C \in \mathcal{C} :\ x \in A \cap C \iff x \in \bigcup_C (A \cap C)$.

(b) $x \in A \setminus \bigcap \mathcal{C} \iff x \in A$ and not$(\forall C : x \in C) \iff x \in A$ and $\exists C : x \notin C \iff \exists C : x \in A \setminus C$ $\iff x \in \bigcup_C (A \setminus C)$. (The key step is negating a quantifier: $\neg\forall = \exists\neg$.) ∎

EXERCISE 1.3 (COUNTING INJECTIONS AND SURJECTIONS)

$S = \{1, 2\}$, $T = \{1, 2, 3\}$. How many (a) injections $S \to T$, (b) surjections $S \to T$, (c) surjections $T \to S$?

▾ Solution

(a) $f(1)$ has 3 choices and $f(2) \neq f(1)$ has 2, giving **6**.

(b) $\#\operatorname{Ran} f \leq \#S = 2 < 3$, so there are **0**.

(c) There are $2^3 = 8$ functions $T \to S$. The non-surjective ones are the 2 constant functions, leaving **6**. (This also illustrates the Corollary: injections $S \to T$ exist and so do surjections $T \to S$.)

EXERCISE 1.4 (INJECTIVITY AND SURJECTIVITY UNDER COMPOSITION)

$f : S \to T$, $g : T \to U$. (a)(i) f, g injective ⇒ $g \circ f$ injective; (ii) $g \circ f$ injective ⇒ f injective. (b)(i) f, g surjective ⇒ $g \circ f$ surjective; (ii) $g \circ f$ surjective ⇒ g surjective. (c)(i) f, g bijective ⇒ $g \circ f$ bijective; (ii) $g \circ f$ bijective ⇒ f injective and g surjective.

▾ Solution

(a)(i) $g(f(s_1)) = g(f(s_2)) \Rightarrow f(s_1) = f(s_2)$ (g injective) $\Rightarrow s_1 = s_2$ (f injective).

(a)(ii) $f(s_1) = f(s_2) \Rightarrow g(f(s_1)) = g(f(s_2)) \Rightarrow s_1 = s_2$, since $g \circ f$ is injective.

(b)(i) Given $u \in U$, there is t with $g(t) = u$ and s with $f(s) = t$, so $(g \circ f)(s) = u$.

(b)(ii) Given u, there is s with $g(f(s)) = u$, so $t := f(s)$ satisfies $g(t) = u$.

(c) (i) follows from (a)(i) and (b)(i). (ii) follows from (a)(ii) and (b)(ii).

Note that $g \circ f$ injective does **not** force g to be injective, and $g \circ f$ surjective does not force f to be surjective. The $\mathbb{N}$ example ($g \circ f = \mathrm{id}$ with $f(n) = n + 1$) shows both.

EXERCISE 1.5 (SIMPLE INJECTIVITY PROBLEM)

10 marks

$f : \mathbb{R} \setminus \{1\} \to \mathbb{R}$, $x \mapsto \frac{2x}{x-1}$. (a) Show f is injective. (b) Find $\operatorname{Ran} f$. (c) Find a left inverse $g : \mathbb{R} \to \mathbb{R} \setminus \{1\}$.

▾ Solution

(a) If $\frac{2x}{x-1} = \frac{2y}{y-1}$ then $2x(y-1) = 2y(x-1)$, i.e. $2xy - 2x = 2xy - 2y$, so $x = y$.

(b) For $y \in \mathbb{R}$: $y = \frac{2x}{x-1} \iff y(x-1) = 2x \iff x(y-2) = y$. If $y = 2$ this says $0 = 2$, impossible. If $y \neq 2$ then $x = \frac{y}{y-2}$, and $x \neq 1$ (otherwise $y = y - 2$). So $\operatorname{Ran} f = \mathbb{R} \setminus \{2\}$.

(c) Let $g(y) = \frac{y}{y-2}$ for $y \neq 2$, and $g(2) = 0$ (any value in $\mathbb{R} \setminus \{1\}$ works). Check: $f(x) - 2 = \frac{2}{x-1}$, so $g(f(x)) = \frac{2x}{x-1} \cdot \frac{x-1}{2} = x$, i.e. $g \circ f = \mathrm{id}_{\mathbb{R}\setminus\{1\}}$. The value $g(2)$ is arbitrary, which shows left inverses are not unique.

EXERCISE 1.6 ($\mathcal{F}(S;T) \equiv T^S$)

$\phi : P = \prod_{s\in S} T_s \to F = \mathcal{F}(S;T)$, $(t_s)_{s\in S} \mapsto f$ with $f(s) = t_s$. Define an inverse ψ and verify it. You may assume $S, T \neq \emptyset$.

▾ Solution

(a) Define $\psi : F \to P$ by $\psi(f) = (f(s))_{s\in S}$. This lies in P because each $f(s) \in T = T_s$.

(b) $\psi(\phi((t_s)_s)) = \psi(f)$ where $f(s) = t_s$, which equals $(f(s))_s = (t_s)_s$, so $\psi \circ \phi = \mathrm{id}_P$ (equality in P is componentwise). Next, $\phi(\psi(f))$ is the function $s \mapsto f(s)$, which has the same domain, codomain and values as f. So $\phi \circ \psi = \mathrm{id}_F$. Hence ϕ is a bijection with inverse ψ. ∎

EXERCISE 1.7 ($\mathcal{P}(S) \equiv \mathcal{F}(S;\{0,1\})$)

10 marks

$\chi : A \mapsto \mathbf{1}_A$. (a) Define $\sigma : F \to P$. (b) Show $\sigma \circ \chi = \mathrm{id}_P$ and $\chi \circ \sigma = \mathrm{id}_F$.

▾ Solution

(a) $\sigma(g) := g^{-1}(\{1\}) = \{s \in S : g(s) = 1\}$.

(b) $\sigma(\chi(A)) = \{s : \mathbf{1}_A(s) = 1\} = A$. For $g \in F$, let $A = \sigma(g)$. If $s \in A$ then $\mathbf{1}_A(s) = 1 = g(s)$. If $s \notin A$ then $g(s) \neq 1$, so $g(s) = 0 = \mathbf{1}_A(s)$, because g only takes the values 0 and 1. Hence $\chi(\sigma(g)) = g$. ∎

EXERCISE 1.8 (IMAGES OF UNIONS AND INTERSECTIONS)

(a) $f(\bigcup \mathcal{A}) = \bigcup f(A)$; (b) $f(\bigcap \mathcal{A}) \subset \bigcap f(A)$; and, from the text, (c) $f(A \setminus B) \supset f(A) \setminus f(B)$.

▾ Solution

(a) $t \in f(\bigcup \mathcal{A}) \iff \exists s \in \bigcup \mathcal{A}$ with $f(s) = t \iff \exists A \in \mathcal{A}\, \exists s \in A$ with $f(s) = t \iff \exists A : t \in f(A) \iff t \in \bigcup f(A)$.

(b) If $t = f(s)$ with $s \in \bigcap \mathcal{A}$, then $s \in A$ for every A, so $t \in f(A)$ for every A. The reverse inclusion fails in general; see the constant-function example above. Equality holds if f is injective.

(c) Let $t \in f(A) \setminus f(B)$, so $t = f(a)$ with $a \in A$. If $a \in B$ then $t \in f(B)$, a contradiction. So $a \in A \setminus B$ and $t \in f(A \setminus B)$. (Strict inclusion example: f constant, $A = \{0, 1\}$, $B = \{1\}$. Then $f(A \setminus B) = \operatorname{Ran} f$ but $f(A) \setminus f(B) = \emptyset$.) ∎

EXERCISE 1.9 (UNIONS AND COMPLEMENTS UNDER INVERSE IMAGES)

10 marks

Show $f^{-1}(\bigcup \mathcal{C}) = \bigcup f^{-1}(C)$ and $f^{-1}(D \setminus C) = f^{-1}(D) \setminus f^{-1}(C)$; in particular $f^{-1}(T \setminus C) = S \setminus f^{-1}(C)$.

▾ Solution

$s \in f^{-1}(\bigcup \mathcal{C}) \iff f(s) \in \bigcup \mathcal{C} \iff \exists C : f(s) \in C \iff \exists C : s \in f^{-1}(C) \iff s \in \bigcup f^{-1}(C)$.

$s \in f^{-1}(D \setminus C) \iff f(s) \in D$ and $f(s) \notin C \iff s \in f^{-1}(D)$ and $s \notin f^{-1}(C) \iff s \in f^{-1}(D) \setminus f^{-1}(C)$.

Take $D = T$ and use $f^{-1}(T) = S$. ∎

EXERCISE 1.10 (DISJOINT UNION OF EQUIVALENCE RELATIONS)

$A_1 \cap A_2 = \emptyset$, with $\sim_i$ an equivalence relation on A_i. Define $x \sim y$ iff ($x, y \in A_1$ and $x \sim_1 y$) or ($x, y \in A_2$ and $x \sim_2 y$). Show $\sim$ is an equivalence relation on $A_1 \cup A_2$.

▾ Solution

Reflexive: $x \in A_i$ for some i, and $x \sim_i x$. **Symmetric**: if $x, y \in A_i$ with $x \sim_i y$, then $y \sim_i x$, so $y \sim x$.

Transitive: suppose $x \sim y$ (so $x, y \in A_i$ with $x \sim_i y$) and $y \sim z$ (so $y, z \in A_j$ with $y \sim_j z$). Then $y \in A_i \cap A_j$. Since $A_1 \cap A_2 = \emptyset$, this forces $i = j$. By transitivity of $\sim_i$, $x \sim_i z$, so $x \sim z$. (Disjointness is used exactly here.) The resulting partition is the union of the two partitions. ∎

EXERCISE 1.11 (SYMMETRY + TRANSITIVITY ⇏ REFLEXIVITY)

10 marks

(a) Find the logical error in the "proof" that symmetry and transitivity give reflexivity. (b) Give a counterexample.

▾ Solution

(a) The argument says: "for all y with $x \sim y$, …, therefore $x \sim x$". It **assumes such a y exists**. If x is related to nothing, the "for all y" statement is **vacuously true** and gives no information, so $x \sim x$ is never derived. What the argument does prove: if every element is related to *something*, then a symmetric and transitive relation is reflexive.

(b) $S = \{1, 2\}$ with $\sim = \{(1, 1)\}$ (only $1 \sim 1$). It is symmetric and transitive (check all cases), but $2 \not\sim 2$. The empty relation on any nonempty set also works.

EXERCISE 1.12 (COUNTABILITY OF $\mathbb{Z}$)

▾ Solution

Define $f : \mathbb{N} \to \mathbb{Z}$ by $f(n) = n/2$ if n is even and $f(n) = -(n-1)/2$ if n is odd. So $1, 2, 3, 4, 5, \ldots \mapsto 0, 1, -1, 2, -2, \ldots$

It is a bijection, with inverse $g(k) = 2k$ for $k > 0$ and $g(k) = 1 - 2k$ for $k \leq 0$. Check: for $k > 0$, $f(2k) = k$; for $k \leq 0$, $1 - 2k$ is odd and $f(1 - 2k) = -(1 - 2k - 1)/2 = k$. Also $g(f(n)) = n$ in both parity cases. Hence $\mathbb{Z}$ is countable. ∎

EXERCISE 1.13 (POLYNOMIALS WITH RATIONAL COEFFICIENTS)

▾ Solution

Let S_n be the polynomials of degree at most n with rational coefficients. The map $\mathbb{Q}^{n+1} \to S_n$, $(a_0, \ldots, a_n) \mapsto a_0 + a_1 x + \cdots + a_n x^n$, is a **surjection**. $\mathbb{Q}^{n+1}$ is countable as a finite product of countable sets, so S_n is countable. Finally $S = \bigcup_{n=0}^{\infty} S_n$ is a countable union of countable sets, hence countable. ∎

EXERCISE 1.14 ($\mathcal{P}_{00}(\mathbb{N})$ IS COUNTABLE)

▾ Solution

Let $\mathcal{P}^{(N)}(\mathbb{N}) = \{A \subset \mathbb{N} : \#A = N\}$. Then $\mathcal{P}^{(0)} = \{\emptyset\}$ is countable. For $N \geq 1$, list A increasingly as $a_1 < \cdots < a_N$ and send $A \mapsto (a_1, \ldots, a_N) \in \mathbb{N}^N$. This is **injective**, because A is recovered as the set of entries. $\mathbb{N}^N$ is countable, so $\mathcal{P}^{(N)}(\mathbb{N})$ is countable. Then $\mathcal{P}_{00}(\mathbb{N}) = \bigcup_{N=0}^{\infty} \mathcal{P}^{(N)}(\mathbb{N})$ is a countable union of countable sets, hence countable. ∎

Contrast: $\mathcal{P}(\mathbb{N})$ itself is uncountable (Cantor).

EXERCISE 1.15 (TOTAL ORDER ↔ STRICT TOTAL ORDER)

10 marks

▾ Solution

(a) Let $\leq$ be a total order and $s < t :\iff s \leq t,\ s \neq t$.

Trichotomy. If $s = t$, neither $s < t$ nor $t < s$ holds. If $s \neq t$, comparability gives $s \leq t$ or $t \leq s$, so $s < t$ or $t < s$. Both cannot hold, since antisymmetry would give $s = t$. So exactly one holds.

Transitivity. If $r < s$ and $s < t$, then $r \leq t$. If $r = t$ we would have $r \leq s \leq r$, so $r = s$, a contradiction. Hence $r < t$.

(b) Let $<$ be a strict total order and $s \leq t :\iff s < t$ or $s = t$.

Reflexive: $s = s$. **Antisymmetric**: if $s \leq t$, $t \leq s$ and $s \neq t$, then $s < t$ and $t < s$, contradicting trichotomy (exactly one holds). **Transitive**: if any of the relations is an equality the claim is immediate; otherwise use transitivity of $<$. **Comparable**: trichotomy gives $s < t$, $s = t$ or $t < s$. ∎

EXERCISE 1.16 (BIJECTIONS ON A 3-ELEMENT SET)

10 + 10 marks

$f : \{x, y, z\} \to \{a, b, c\}$ bijective. A: $f(x) = a$. B: $f(y) \neq a$. C: $f(z) \neq b$.

▾ Solution

There are $3! = 6$ bijections. Write them as $(f(x), f(y), f(z))$:

f	A	B	C	#true
(a,b,c)	T	T	T	3
(a,c,b)	T	T	F	2
(b,a,c)	F	F	T	1
(b,c,a)	F	T	T	2
(c,a,b)	F	F	F	0
(c,b,a)	F	T	T	2

(a) Exactly two true: $x \mapsto a, y \mapsto c, z \mapsto b$; $x \mapsto b, y \mapsto c, z \mapsto a$; $x \mapsto c, y \mapsto b, z \mapsto a$.

(b) Exactly one true: $x \mapsto b, y \mapsto a, z \mapsto c$.

EXERCISE 1.17 (RELATIONS VIA FUNCTIONS & PARTITIONS)

20 marks

$f : S \to T$, $\mathcal{C}$ a partition of S. $t_1 \sim t_2$ iff $\exists A \in \mathcal{C} :\ f^{-1}(\{t_1, t_2\}) \subset A$. (a) Reflexive? If f is injective? (b) Symmetric? If f is bijective? (c) Transitive? If f is surjective?

▾ Solution

(a) **Not necessarily reflexive.** $t \sim t$ iff the fibre $f^{-1}(\{t\})$ lies inside one block. Counterexample: $S = \{1, 2\}$, $\mathcal{C} = \{\{1\}, \{2\}\}$, $T = \{t\}$, f constant. Then $f^{-1}(\{t\}) = \{1, 2\}$ is in no single block, so $t \not\sim t$.
If f is injective, it is reflexive: $f^{-1}(\{t\})$ has at most one point. If it is empty, it lies in any block ($\mathcal{C} \neq \emptyset$ since $S \neq \emptyset$). If it is $\{s\}$, then s lies in some block, since the blocks cover S.

(b) **Always symmetric**, with no hypothesis on f, since $\{t_1, t_2\} = \{t_2, t_1\}$. In particular it is symmetric when f is bijective.

(c) **Not necessarily transitive.** $S = \{1, 2\}$, $\mathcal{C} = \{\{1\}, \{2\}\}$, $T = \{a, b, c\}$, $f(1) = a$, $f(2) = c$. Then $f^{-1}\{a, b\} = \{1\}$, so $a \sim b$; $f^{-1}\{b, c\} = \{2\}$, so $b \sim c$; but $f^{-1}\{a, c\} = \{1, 2\}$, so $a \not\sim c$. The value b is missed by f, and that is what breaks transitivity.
If f is surjective, it is transitive: let $t_1 \sim t_2$ via A and $t_2 \sim t_3$ via B. Since f is surjective, $\emptyset \neq f^{-1}(\{t_2\}) \subset A \cap B$. Distinct blocks are disjoint, so $A = B$. Then $f^{-1}\{t_1, t_3\} \subset f^{-1}\{t_1, t_2\} \cup f^{-1}\{t_2, t_3\} \subset A$, so $t_1 \sim t_3$.

So if f is bijective, $\sim$ is an equivalence relation, and its partition is $\{f(A) : A \in \mathcal{C}\}$.

EXERCISE 1.18 (UNCOUNTABILITY OF $\mathbb{R}$)

10 marks

(a) $\psi : \{0, 1\}^{\mathbb{N}} \to \mathbb{R}$, $a \mapsto \sum_{n \geq 1} a_n 10^{-n}$ is injective. (b) Deduce that $\mathbb{R}$ is uncountable.

▾ Solution

(a) The series converges by comparison with $\sum 10^{-n}$. Let $a \neq b$ and let m be the **least** index with $a_m \neq b_m$ (well-ordering); WLOG $a_m = 1$, $b_m = 0$. Then

$$\psi(a) - \psi(b) = 10^{-m} + \sum_{n>m} (a_n - b_n) 10^{-n} \ \geq\ 10^{-m} - \sum_{n>m} 10^{-n} = 10^{-m} - \tfrac{1}{9}\, 10^{-m} = \tfrac{8}{9}\, 10^{-m} > 0.$$

So $\psi(a) \neq \psi(b)$.

(b) Suppose $\mathbb{R}$ were countable, with injection $j : \mathbb{R} \to \mathbb{N}$. Let $\beta : \mathcal{P}(\mathbb{N}) \to \mathcal{F}(\mathbb{N}; \{0, 1\}) \to \{0, 1\}^{\mathbb{N}}$ be the bijection from Exercises 1.6–1.7. Then $j \circ \psi \circ \beta : \mathcal{P}(\mathbb{N}) \to \mathbb{N}$ is an injection (a composition of injections), contradicting **Cantor's Theorem** (no injection $\mathcal{P}(S) \to S$). Hence $\mathbb{R}$ is uncountable. ∎

EXERCISE 1.19 (CHARACTERISATION OF INFINITENESS)

10 marks

S is infinite $\iff$ there is an injection $\mathbb{N} \to S$.

▾ Solution

($\Leftarrow$) Suppose $\iota : \mathbb{N} \to S$ is injective but S is finite. If $S = \emptyset$ there is no map from $\mathbb{N}$ at all. Otherwise there is a bijection $\beta : S \to \{1, \dots, K\}$, and $\beta \circ \iota$ restricted to $\{1, \dots, K+1\}$ is an injection of a $(K+1)$-element set into a K-element set, contradicting the **pigeonhole principle**.

($\Rightarrow$) Let S be infinite. By AC, take a choice function c on $\mathcal{P}(S) \setminus \{\emptyset\}$. Define recursively $s_1 = c(S)$ and $s_{n+1} = c(S \setminus \{s_1, \dots, s_n\})$. The set $S \setminus \{s_1, \dots, s_n\}$ is nonempty, since otherwise S would be finite. Then $s_{n+1} \notin \{s_1, \dots, s_n\}$, so $n \mapsto s_n$ is injective. ∎

IN-TEXT EXERCISE (COUNTABILITY REMARK 2)

For $S \neq \emptyset$: countable $\iff$ there is a surjection $\mathbb{N} \to S$.

▾ Solution

An injection $f : S \to \mathbb{N}$ has a left inverse $g : \mathbb{N} \to S$ (here $S \neq \emptyset$ is needed), and g is surjective because $g \circ f = \mathrm{id}_S$ (Exercise 1.4(b)(ii)). Conversely, a surjection $g : \mathbb{N} \to S$ has a right inverse h (by AC; or explicitly, $h(s) = \min g^{-1}(\{s\})$ by well-ordering), and h is injective. ∎

Section 2: Real analysis & linear algebra (revision)

The engine room. The LUB Principle, Approximation of suprema, monotone convergence, Bolzano–Weierstrass, the Uniform Continuity Theorem, norms and inner products. Later sections quote these constantly.

2.1 Subsets of $\mathbb{R}$

MINIMUM, MAXIMUM, BOUNDS

A has a **minimum** if $\exists x \in A\ \forall a \in A:\ a \geq x$. The **maximum** is defined similarly. When they exist they are unique, written $\min A$ and $\max A$. Also $\max A = -\min(-A)$, where $-A = \{-a : a \in A\}$.

Bounded above: $\exists x\ \forall a:\ a \leq x$, and x is an **upper bound**. **Bounded below** is similar. **Bounded**: $\exists D\ \forall a, b \in A:\ |b - a| \leq D$, equivalently $A \subset [x, y]$ for some x, y.

The set of upper bounds U_A is "upward closed" ($x \in U_A$ and $y > x$ imply $y \in U_A$). If $\max A$ exists then $U_A = [\max A, \infty[$.

Examples: a nonempty finite set has a min and a max. $[a, b[$ has a min but **no max**; this obvious-looking fact depends on the Archimedean property.

THEOREM (WELL-ORDERING OF $\mathbb{N}$)

Every nonempty $A \subset \mathbb{N}$ has a minimum.

Proof. Suppose A has no minimum. Prove by **induction** that $P_n : \{1, \ldots, n\} \subset A^{\complement}$. P_1: if $1 \in A$ it would be the min. $P_k \Rightarrow P_{k+1}$: if $k+1 \in A$ it would be the min, since $1, \ldots, k \notin A$. So $A^{\complement} = \bigcup_n \{1, \ldots, n\} = \mathbb{N}$, i.e. $A = \emptyset$, a contradiction. ∎

THE NINE TYPES OF SUBINTERVAL OF $\mathbb{R}$

Take $a \leq b$ and $c < d$.

I. Closed: $]-\infty, b]$, $[a, b]$, $[a, \infty[$.

II. Open: $]-\infty, d[$, $]c, d[$, $]c, \infty[$.

III. Half-open: $[c, d[$, $]c, d]$.

IV. Both open and closed: $]-\infty, \infty[= \mathbb{R}$.

If $a = b$, $[a, b] = \{a\}$ is **degenerate**. The bounded types are I(ii), II(ii), III(i), III(ii). $I = [0, 1]$ is the **unit interval**.

Germ of "closedness": if $x_n \in J$ for a closed interval J and $x_n \to x$, then $x \in J$.

Germ of "openness": if $p \in J$ for an open interval J, then $]p - \delta, p + \delta[\subset J$ for some $\delta > 0$.

ARCHIMEDEAN PROPERTY: FOUR EQUIVALENT FORMS

(i) $\mathbb{Q}$ is **order dense** in $\mathbb{R}$: $x < y \Rightarrow \exists q \in \mathbb{Q} :\ x < q < y$.

(ii) $\forall x > 0\ \exists n \in \mathbb{N} :\ 1/n < x$.

(iii) $\forall x > 0\ \forall y \in \mathbb{R}\ \exists n :\ nx \geq y$.

(iv) $\mathbb{N}$ is unbounded in $\mathbb{R}$: $\forall x\ \exists n :\ n > x$.

Proof.

(ii)⇒(i), case $y > 0$: pick n with $1/n < y - x$. By (ii) the set $\{k \in \mathbb{N} : k/2n \geq y\}$ is nonempty; let p be its minimum (well-ordering). Then $q = (p-1)/2n$ satisfies $q < y$ and $q = p/2n - 1/2n \geq y - 1/2n > y - 1/n > x$. Case $y \leq 0$: apply the first case to $-y < -x$ (note $-x > 0$) and negate.

(i)⇒(iv): if $x < 1$ take $n = 1$. If $x \geq 1$, (i) gives a rational $m/n \in]x, x + 1[$ with $m, n \in \mathbb{N}$, and then $m \geq m/n > x$.

(iv)⇒(iii): choose $n > y/x$, so $nx > y$.

(iii)⇒(ii): apply (iii) with $y = 2$: $nx \geq 2 > 1$, so $1/n < x$. ∎

LEAST UPPER BOUND PRINCIPLE (AN AXIOM OF $\mathbb{R}$)

Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound $\sup A$, the **supremum**. Equivalently, $U_A = [s_A, \infty[$ and $\sup A = \min U_A$.

It is **not a theorem**: it is part of what we mean by $\mathbb{R}$ (the complete ordered field). From it we derive (1) the Archimedean property of $\mathbb{R}$ and (2) Cauchy completeness of $\mathbb{R}$.

APPROXIMATION OF SUPREMA (USED EVERYWHERE)

If $x = \sup A$, then for all $\varepsilon > 0$ there is $a \in A$ with $x - \varepsilon < a \leq x$, i.e. $]x - \varepsilon, x] \cap A \neq \emptyset$. The infimum version is analogous.

THEOREM (ARCHIMEDEAN PROPERTY FOR $\mathbb{R}$)

Proof. Suppose $\mathbb{N}$ is bounded above, with $s = \sup \mathbb{N}$. Then $s - 1$ is not an upper bound, so $N > s - 1$ for some $N \in \mathbb{N}$. But then $N + 1 > s$ and $N + 1 \in \mathbb{N}$, a contradiction. So version (iv) holds. ∎

Consequence: $(1/n)$ is a nonconstant convergent sequence. Also $[a, b[$ and $[0, 2\pi] \cap \mathbb{Q}$ have no maximum: given x in the set, density gives a rational q with $x < q < b$ (resp. $< 2\pi$), and q lies in the set. For the second set this uses $2\pi \notin \mathbb{Q}$.

STRANGE BUT TRUE

$\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ "interlace": each is order dense in $\mathbb{R}$. Yet $\mathbb{Q}$ is countable and $\mathbb{R} \setminus \mathbb{Q}$ is uncountable.

2.2 Sequences in $\mathbb{R}$

SEQUENCE DEFINITIONS

A real sequence is an element of $\mathbb{R}^{\mathbb{N}} \leftrightarrow \mathcal{F}(\mathbb{N}; \mathbb{R})$, a natural linear isomorphism. The **set of terms** is $X = \{x_n : n \in \mathbb{N}\}$, a countable nonempty set. Properties:

- **Bounded above/below/bounded**: X is. We write $\sup x_n := \sup X$.
- **Monotonic increasing**: $x_{n+1} \geq x_n$ for all n. **Monotonic decreasing**: $x_{n+1} \leq x_n$.
- **Convergent**: $\exists a\ \forall \varepsilon > 0\ \exists N\ \forall n \geq N :\ |x_n - a| < \varepsilon$.
- **Cauchy**: $\forall \varepsilon > 0\ \exists N\ \forall n, m \geq N :\ |x_n - x_m| < \varepsilon$.

Convergence depends only on the **tail**: (x_n) converges iff $(x_{p+n})_n$ does, with the same limit.

PROPOSITION (APPROXIMATION OF SUPREMA BY SEQUENCES)

If $A \neq \emptyset$ is bounded above, there is a **monotonic increasing** sequence (a_n) in A with $a_n \to \sup A$. If $\sup A \notin A$, it can be chosen strictly increasing.

Proof. If $s := \sup A \in A$, use the constant sequence. Otherwise choose $a_1 \in]s - 1, s] \cap A$, and given $a_1 < \cdots < a_n < s$ choose $a_{n+1} \in A$ with $\max\{a_n, s - \frac{1}{n+1}\} < a_{n+1} < s$, by Approximation of suprema. Then $|s - a_n| < 1/n \to 0$ by the Archimedean property and the Sandwich Principle. ∎

PROPOSITION (ALGEBRA OF LIMITS IN $\mathbb{R}$)

If $x_n \to a$ and $y_n \to b$:

(1) $sx_n + ty_n \to sa + tb$; (2) $x_n y_n \to ab$; (3) $|x_n| \to |a|$; (4) if $a \neq 0$ then $x_n \neq 0$ eventually and $1/x_n \to 1/a$;

(5a) if $x_n \leq y_n$ eventually then $a \leq b$ **(order preservation: ≤ survives limits)**; (5b) if $a < b$ then $x_n < y_n$ eventually;

(6) **Sandwich Principle**: if $a = b$ and $x_n \leq z_n \leq y_n$ eventually, then $z_n \to a$. This is the most-used convergence criterion.

PITFALL

Strict inequality does **not** survive limits: $1/n > 0$ for all n, but $\lim 1/n = 0$. So $(1/n)$ lies in $]0, 1]$ while its limit does not. The interval $]0, 1]$ is not closed.

THEOREM (BOUNDED MONOTONIC SEQUENCES CONVERGE)

If (x_n) is increasing and bounded above, then $x_n \to \sup\{x_n\}$. (If decreasing and bounded below, then $x_n \to \inf$.)

Proof. Let $s = \sup x_n$ and $\varepsilon > 0$. By Approximation of suprema, $x_N > s - \varepsilon$ for some N. By monotonicity, $s - \varepsilon < x_N \leq x_n \leq s$ for $n \geq N$, so $|s - x_n| < \varepsilon$. This rests on the LUB Principle. Decreasing case: apply the result to $(-x_n)$ and use $\inf A = -\sup(-A)$. ∎

SUBSEQUENCE

$(x_{n_k})_k$ with $n_1 < n_2 < \cdots$, i.e. given by an order-preserving injection $k \mapsto n_k$. A subsequence of a subsequence is a subsequence. Example: $(1, 1, \dots)$ and $(-1, -1, \dots)$ are subsequences of $((-1)^n)$.

THEOREM (EVERY REAL SEQUENCE HAS A MONOTONIC SUBSEQUENCE)

Proof. Call n a **peak index** if $x_{n+p} \leq x_n$ for all $p \in \mathbb{N}$, and let P be the set of peak indices. **Case I**: P infinite. List P increasingly as $n_1 < n_2 < \cdots$; then (x_{n_i}) is decreasing. **Case II**: P finite. Start at n_1 beyond every peak. n_1 is not a peak, so some $n_2 > n_1$ has $x_{n_2} > x_{n_1}$; n_2 is not a peak either, and so on. This gives an increasing subsequence. ∎

THEOREM (BOLZANO–WEIERSTRASS, BWT)

Every sequence in a **closed and bounded interval** J has a subsequence converging to a limit **in** J.

Proof. Take a monotonic subsequence. It is bounded and monotonic, hence convergent, and its limit is in J because J is closed. ∎ (The classical proof bisects J repeatedly, keeping a half that contains infinitely many terms. That gives a Cauchy subsequence, which converges by completeness.)

LEMMA (SUBSEQUENCE LEMMA)

TFAE: (i) some subsequence converges to a; (ii) for all $\varepsilon > 0$, $\{n : |x_n - a| < \varepsilon\}$ is infinite.

Proof. (i)⇒(ii): the set contains the tail $\{n_K, n_{K+1}, \ldots\}$. (ii)⇒(i): pick $n_{k+1} > n_k$ in $\{n : |x_n - a| < \frac{1}{k+1}\}$, which is possible because each such set is infinite. Then $|x_{n_k} - a| < 1/k \to 0$. ∎

LIM SUP AND LIM INF

For bounded (x_n), let $u_n = \sup_{k \geq n} x_k$ and $l_n = \inf_{k \geq n} x_k$. Then $\inf X = l_1 \leq l_2 \leq \cdots \leq u_2 \leq u_1 = \sup X$, so both sequences are monotone and bounded, hence convergent:

$$\limsup_{n \to \infty} x_n := \lim_{n} \sup_{k \geq n} x_k, \qquad \liminf_{n \to \infty} x_n := \lim_{n} \inf_{k \geq n} x_k.$$

Proposition. If (x_n) is bounded above and $\limsup x_n$ exists, some subsequence converges to $\limsup x_n$.

THEOREM (COMPLETENESS OF $\mathbb{R}$)

TFAE: (i) (x_n) converges; (ii) (x_n) is **Cauchy**; (iii) (x_n) is bounded and $\liminf x_n = \limsup x_n$. In that case $\lim x_n = \limsup x_n = \liminf x_n$.

Proof.

(i)⇒(ii): $|x_n - x_m| \leq |x_n - a| + |a - x_m| < 2\varepsilon$. This holds in every metric space.

(ii)⇒(iii): with $\varepsilon = 1$, $|x_n| \leq |x_N| + 1$ for $n \geq N$, so $|x_n| \leq \max\{|x_1|, \ldots, |x_{N-1}|, |x_N| + 1\}$ and the sequence is bounded. Then pick $n, m \geq N$ with $|x_n - l| < \varepsilon$ and $|u - x_m| < \varepsilon$ (possible because l, u are limits of subsequences). This gives $|u - l| < 3\varepsilon$, so $u = l$.

(iii)⇒(i): $l_n \leq x_n \leq u_n$ and both bounds tend to a. Sandwich. ∎

2.3 Linear algebra

VECTOR SPACE OVER A FIELD $\mathbb{K}$

$(V, +)$ is an **abelian group** (identity 0_V, inverses, associativity, commutativity). Scalar multiplication satisfies $1v = v$ and $(\lambda\mu)v = \lambda(\mu v)$. Two distributive laws: $\lambda(u+v) = \lambda u + \lambda v$ and $(\lambda+\mu)v = \lambda v + \mu v$.

Consequences: the identity is unique, inverses are unique, $(-\lambda)v = -(\lambda v)$, $0v = 0$ and $\lambda 0 = 0$.

Examples: $\{0\}$; $\mathbb{K}^N$; $M_{n,m}(\mathbb{K})$; $\mathcal{F}(S;V)$ with pointwise operations (zero = constant function 0_V); the sequence space $s(V)$ (zero = constant zero sequence).

LINEAR MAPS, DUAL, SUBSPACES

T is **linear** if $T(u+\lambda v) = Tu + \lambda Tv$. A bijective linear map is a **linear isomorphism**. The **dual** V^{dual} is the set of linear maps $V \to \mathbb{K}$. Every $\phi \in (\mathbb{R}^N)^{\text{dual}}$ has the form $\phi(x) = a \cdot x$ with $a_i = \phi(e_i)$ (where e_i are the standard basis vectors).

A **(linear) subspace** U satisfies (VSS.1) $0 \in U$ (or just $U \neq \emptyset$) and (VSS.2) $u + \lambda u' \in U$. A subspace is itself a vector space. The most important "mother space" is $\mathcal{F}(S;V)$.

$M_A : x \mapsto Ax$ is linear, and $A \mapsto M_A$ is a linear isomorphism $M_{n,m}(\mathbb{K}) \to L(\mathbb{K}^m;\mathbb{K}^n)$. Evaluation maps $f \mapsto f(s_0)$ are linear. $\operatorname{Ker} T = T^{-1}(\{0\})$ and $\operatorname{Ran} T = T(U)$ are subspaces.

NORM

A **norm** on V (over $\mathbb{R}$ or $\mathbb{C}$) is $\|\cdot\| : V \to [0,\infty[$ with

(N1) $\|v\| = 0 \Rightarrow v = 0$ **(Definiteness)**;

(N2) $\|\lambda v\| = |\lambda|\|v\|$ **(Homogeneity)**;

(N3) $\|u+v\| \leq \|u\| + \|v\|$ **(Triangle inequality)**.

(N2) gives $\|0\| = 0$. Examples: $\|x\|_1 = \sum |x_i|$; $\|x\|_2$; $\|x\|_\infty = \max |x_i|$; $\|f\|_1 = \int_J |f|$ on $C(J;\mathbb{R})$ (**not quite** a norm on $\mathcal{R}(J;\mathbb{R})$, because (N1) fails); $\|f\|_{\sup} = \sup_J |f|$. Also $\|f\|_1 \leq |J|\,\|f\|_{\sup}$.

INNER PRODUCT

$\langle\cdot,\cdot\rangle : V \times V \to \mathbb{F}$ with (IP.1) $\langle v,v\rangle > 0$ for $v \neq 0$; (IP.2) $\langle v,u\rangle = \overline{\langle u,v\rangle}$; (IP.3) linear in the second slot.

Examples: $\sum \bar{u}_i v_i$ on $\mathbb{F}^N$; $\int_J fg$ on $C(J;\mathbb{R})$.

Cauchy–Schwarz: $|\langle u,v\rangle|^2 \leq \langle u,u\rangle\langle v,v\rangle$, with equality iff u, v are linearly dependent.

$\|v\| = \sqrt{\langle v,v\rangle}$ is an **inner product norm**. $\|\cdot\|_{\sup}$ is **not** one; see the in-text questions below.

2.4 Continuity and boundedness

DEFINITIONS (FOR $F : A \to \mathbb{R}$, $A \subset \mathbb{R}$)

- **Bounded**: Ran f is bounded. Then $\|f\|_{\sup} = \sup |f|$ exists, by LUB.
- **Continuous at p**: $\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A:\ |x - p| < \delta \Rightarrow |f(x) - f(p)| < \varepsilon$.
- **Not continuous at p**: $\exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in A:\ |x - p| < \delta$ but $|f(x) - f(p)| \geq \varepsilon$.
- **Sequentially continuous at p**: $a_n \to p \Rightarrow f(a_n) \to f(p)$.
- **Continuous**: $\forall \varepsilon\ \forall x\ \exists \delta\ \forall y\ \ldots$ (δ may depend on x).
- **Uniformly continuous**: $\forall \varepsilon\ \exists \delta\ \forall x, y\ \ldots$ (one δ for all points).
- **Lipschitz**: $\exists M \geq 0\ \forall x, y:\ |f(x) - f(y)| \leq M|x - y|$.

Lipschitz ⇒ uniformly continuous ⇒ continuous. $C_{\text{Lip}}(A; \mathbb{R}) \subset C(A; \mathbb{R}) \subset \mathcal{F}(A; \mathbb{R})$ and $\mathcal{F}_b(A; \mathbb{R}) \subset \mathcal{F}(A; \mathbb{R})$, all subspaces.

Examples: $1/x$ on $]0, 1]$ is **not uniformly continuous**: the points $\frac{1}{n+1}, \frac{1}{n}$ are $\frac{1}{n(n+1)}$ apart, but their images are 1 apart. $1/x$ on $[1, \infty[$ is **Lipschitz**: $|\frac{1}{x} - \frac{1}{y}| = \frac{|y-x|}{xy} \leq |y - x|$.

THEOREM (UNIFORM CONTINUITY THEOREM)

If $f : J \to \mathbb{R}$ is continuous and J is **closed and bounded**, then (a) f is uniformly continuous, (b) f is bounded, and (c) f attains a maximum and a minimum. In particular $C(J; \mathbb{R}) \subset \mathcal{F}_b(J; \mathbb{R})$.

Proof.

(a) Suppose not. Then there is $\varepsilon > 0$ and points with $|x_n - y_n| < 1/n$ but $|f(x_n) - f(y_n)| \geq \varepsilon$. By **BWT**, $x_{n_k} \to p \in J$, and then $y_{n_k} \to p$ too. By sequential continuity $f(x_{n_k}) - f(y_{n_k}) \to 0$, a contradiction.

(b) If unbounded, pick $|f(x_n)| \geq n$. By BWT $x_{n_k} \to p$, and $f(x_{n_k}) \to f(p)$ contradicts $|f(x_{n_k})| \geq n_k \to \infty$.

(c) Let $u = \sup f(J)$. By Approximation of suprema by sequences, $f(x_n) \to u$. By BWT $x_{n_k} \to p \in J$, so $f(p) = u$. For the min, apply this to $-f$. ∎

INTERMEDIATE VALUE THEOREM (IVT)

If $f : [a, b] \to \mathbb{R}$ is continuous and p lies between $f(a)$ and $f(b)$, then $f(t) = p$ for some $t \in [a, b]$.

Proof. WLOG $f(a) < p < f(b)$ (otherwise use $-f$, or a trivial endpoint case). Let $t = \inf A$ where $A = \{x \in J : f(x) \geq p\}$. A sequence in A converges to t, so $t \in J$ (closed) and $f(t) \geq p$ by continuity; in particular $t > a$. If $f(t) > p$, continuity gives $\delta < t - a$ with $f > p$ on $]t - \delta, t + \delta[\cap J$. Then $t - \delta/2 \in A$, contradicting $t = \inf A$. So $f(t) = p$. ∎

Corollary. The range of a continuous f on $[a, b]$ is a closed bounded interval $[\min f, \max f]$.

2.5 Integration

FACTS ABOUT RIEMANN INTEGRATION ON $J = [A, B]$

$C(J;\mathbb{R}) \subset \mathcal{R}(J) \subset \mathcal{F}_b(J;\mathbb{R})$, all subspaces of $\mathcal{F}(J;\mathbb{R})$.

- **Linearity**: $\int(\lambda f + \mu g) = \lambda \int f + \mu \int g$.
- **Order preservation**: $f \le g \Rightarrow \int f \le \int g$.
- **Fundamental integration inequalities**: $|\int f| \le \int |f| \le |J|\,\|f\|_{\sup}$.

FTC I: if $F' \in C(J;\mathbb{R})$ then $\int_s^t F' = F(t) - F(s)$. **FTC II**: if $f \in C(J;\mathbb{R})$ then $F(t) = \int_a^t f$ is differentiable with $F' = f$.

For a **nondegenerate** J: $f \in C(J;\mathbb{R})$ and $\int |f| = 0 \Rightarrow f = 0$ (Exercise 2.10).

Exercises for Section 2, with full solutions

EXERCISE 2.1 (MAX AND MIN FOR SUBSETS OF $\mathbb{Z}$)

10 + 5 marks

▾ Solution

(a) Let $A \subset \mathbb{Z}$ be nonempty with $a \ge p \in \mathbb{Z}$ for all $a \in A$. Set $A' = \{a - p + 1 : a \in A\}$. Then $A' \subset \mathbb{N}$ (since $a - p + 1 \ge 1$) and $A' \neq \emptyset$. By **well-ordering of $\mathbb{N}$**, A' has a minimum $m = a_0 - p + 1$ for some $a_0 \in A$. For any $a \in A$, $a - p + 1 \ge m$ gives $a \ge a_0$. So $a_0 = \min A$.

(b) If A is bounded above by q, then $-A$ is nonempty and bounded below by $-q$, so $x = \min(-A)$ exists by (a). Then $-x \in A$, and for every $a \in A$, $-a \ge x$, i.e. $a \le -x$. So $\max A = -x = -\min(-A)$. ∎

EXERCISE 2.2 (BOUNDEDNESS BY MODULUS)

10 marks

TFAE: (i) A is bounded above and below; (ii) $\exists M\ \forall a \in A:\ |a| \le M$; (iii) A is bounded.

▾ Solution

(i)⇒(ii): if $l \le a \le u$ for all a, then $|a| \le \max\{|l|, |u|\} =: M$.

(ii)⇒(iii): $|b - a| \le |a| + |b| \le 2M =: D$.

(iii)⇒(i): if $A = \emptyset$, every real number is a bound. Otherwise fix $a_0 \in A$. For all a, $|a - a_0| \le D$, so $a_0 - D \le a \le a_0 + D$. ∎

EXERCISE 2.3 (APPROXIMATION OF SUPREMA)

5 marks

▾ Solution

Let $x = \sup A$ and $\varepsilon > 0$. Since $x - \varepsilon < x$ and x is the *least* upper bound, $x - \varepsilon$ is not an upper bound. So there is $a \in A$ with $a > x - \varepsilon$. Also $a \leq x$, because x is an upper bound. Hence $a \in]x - \varepsilon, x] \cap A$. ∎

EXERCISE 2.4 (HALVING THE WORK: INFIMA)

5 + 5 marks

▾ Solution

(a) x is a lower bound for $A \iff \forall a:\ x \leq a \iff \forall a:\ -a \leq -x \iff -x$ is an upper bound for $-A$.

(b) If A (nonempty) is bounded below by x, then $-A$ is nonempty and bounded above by $-x$. By **LUB**, $s = \sup(-A)$ exists. By (a), $-s$ is a lower bound for A. If l is any lower bound for A, then $-l$ is an upper bound for $-A$, so $s \leq -l$, i.e. $l \leq -s$. Hence $-s$ is the greatest lower bound: $\inf A = -\sup(-A)$. ∎

EXERCISE 2.5 (ORDER DENSITY OF $\mathbb{R}\setminus\mathbb{Q}$)

10 marks

▾ Solution

Use one positive irrational, $\sqrt{2}$. Let $x < y$. Then $x/\sqrt{2} < y/\sqrt{2}$, so by order density of $\mathbb{Q}$ there is $q \in \mathbb{Q}$ with $x/\sqrt{2} < q < y/\sqrt{2}$. If $q = 0$, use density again to replace q by a rational $q' \in]0, y/\sqrt{2}[$; this interval is nonempty since $y > 0$ in that case. So we may take $q \neq 0$. Then $z = q\sqrt{2}$ is irrational (if it were rational, $\sqrt{2} = z/q$ would be too), and $x < z < y$. ∎

EXERCISE 2.6 ($\sup \mathbb{Q} \cap [0, 2\pi] = 2\pi$)

10 marks

▾ Solution

Let $S = \mathbb{Q} \cap [0, 2\pi]$. It is nonempty ($0 \in S$) and 2π is an upper bound. Suppose $u < 2\pi$ were an upper bound. Let $u' = \max\{u, 0\} < 2\pi$. By order density there is $q \in \mathbb{Q}$ with $u' < q < 2\pi$. Then $q \in S$ and $q > u$, a contradiction. So 2π is the least upper bound. (It is not a maximum, since $2\pi \notin \mathbb{Q}$.) ∎

EXERCISE 2.7 (UNIQUENESS OF LIMITS)

10 marks

▾ Solution

Suppose $x_n \to a$, $x_n \to b$ and $a \neq b$. Let $\varepsilon = |a - b|/2 > 0$. There is N_1 with $|x_n - a| < \varepsilon$ for $n \geq N_1$, and N_2 with $|x_n - b| < \varepsilon$ for $n \geq N_2$. For $n = \max\{N_1, N_2\}$: $|a - b| \leq |a - x_n| + |x_n - b| < 2\varepsilon = |a - b|$, a contradiction. ∎

EXERCISE 2.8 (AREA OF UNIT DISC)

20 marks

$a_n =$ area of the regular n-gon inscribed in the unit disc. Find $\sup a_n$.

▾ Solution

D_n splits into n congruent isosceles triangles with two sides of length 1 and apex angle $2\pi/n$. Each has area $\frac{1}{2}\sin\frac{2\pi}{n}$, so

$$a_n = \frac{n}{2}\sin\frac{2\pi}{n} = \pi \cdot \frac{\sin t_n}{t_n}, \qquad t_n = \frac{2\pi}{n}.$$

Monotone. Let $g(t) = \frac{\sin t}{t}$ on $]0, \pi]$. Then $g'(t) = \frac{t\cos t - \sin t}{t^2} < 0$: on $]0, \pi/2[$ because $\tan t > t$, and on $[\pi/2, \pi]$ because $\cos t \leq 0 < \sin t$ (at $t = \pi$ the numerator is $-\pi < 0$). So g is decreasing. Since t_n decreases in n, a_n increases. It is bounded by 4 (as $D \subset [-1, 1]^2$). By the **bounded monotone** theorem, (a_n) converges and $\lim a_n = \sup a_n$.

Limit. By L'Hôpital's rule, $\lim_{t\to 0}\frac{\sin t}{t} = \lim\frac{\cos t}{1} = 1$. Since $t_n \to 0$, $a_n \to \pi$.

Hence $\sup a_n = \pi$, which is the area of the disc. ∎

EXERCISE 2.9 (CONTINUITY ≡ SEQUENTIAL CONTINUITY, IN $\mathbb{R}$)

20 marks

▾ Solution

(a)⇒(b): let $a_n \to p$ and $\varepsilon > 0$. Choose δ from continuity, then N with $|a_n - p| < \delta$ for $n \geq N$. Then $|f(a_n) - f(p)| < \varepsilon$ for $n \geq N$.

(b)⇒(a) by the contrapositive: if f is not continuous at p, there is $\varepsilon_0 > 0$ such that for each $\delta = 1/n$ there is $x_n \in A$ with $|x_n - p| < 1/n$ but $|f(x_n) - f(p)| \geq \varepsilon_0$. Then $x_n \to p$ (Sandwich) but $f(x_n) \not\to f(p)$. So f is not sequentially continuous. ∎

EXERCISE 2.10 (CONTINUOUS F WITH $\int |F| = 0$ IS ZERO)

5 + 10 marks

▾ Solution

(a) On $J = [a, b]$ with $a < b$, take $f = \mathbf{1}_{\{a\}}$. It is Riemann integrable (one discontinuity) with $\int |f| = 0$, but $f \neq 0$.

(b) This is the same argument as Exercise 0.2. Suppose $f(p) \neq 0$ and let $c = |f(p)|$. Continuity gives $\delta > 0$ with $|f| > c/2$ on $[p - \delta, p + \delta] \cap J$, an interval of positive length ℓ because J is nondegenerate. By order preservation, $\int_J |f| \geq \frac{c}{2}\ell > 0$, a contradiction. ∎

IN-TEXT QUESTIONS (LINEAR ALGEBRA AND ANALYSIS)

▾ Answers

- **Inverse of a bijective linear map is linear**: given u, v, let $a = T^{-1}u$ and $b = T^{-1}v$. Then $T(a + \lambda b) = u + \lambda v$, so $T^{-1}(u + \lambda v) = a + \lambda b = T^{-1}u + \lambda T^{-1}v$.
- **Which constant maps are linear?** Only the zero map, since linearity forces $T0 = 0$.
- **Translation T_x is linear** iff $x = 0$, since $T_x(0) = x$.
- **Dilation D_λ** is always linear.
- **$L(U; V)$ is a subspace of $\mathcal{F}(U; V)$**: the zero map is linear, and if S, T are linear then $(S + \lambda T)(u + \mu v) = (S + \lambda T)u + \mu(S + \lambda T)v$.
- **Dual formula**: $a_i = \phi(e_i)$, since $\phi(x) = \phi(\sum x_i e_i) = \sum x_i \phi(e_i)$.
- **$D : p \mapsto p'$ on $P(\mathbb{R})$**: $\operatorname{Ker} D$ is the constant polynomials. $\operatorname{Ran} D = P(\mathbb{R})$, since every polynomial has a polynomial antiderivative.
- **Range of $\rho : f \mapsto \int_J f$**: all of $\mathbb{R}$ when J is nondegenerate, since $\int_J c = c|J|$. It is $\{0\}$ if J is degenerate.
- $\|f\|_1 \leq |J| \|f\|_{\sup}$: $|f| \leq \|f\|_{\sup}$ pointwise, so integrate (order preservation).
- **Why is $\|\cdot\|_{\sup}$ not an inner product norm?** Inner product norms satisfy the **parallelogram law** $\|f + g\|^2 + \|f - g\|^2 = 2\|f\|^2 + 2\|g\|^2$. On $[0, 1]$ with $f = 1$ and $g(x) = x$: the left side is $2^2 + 1^2 = 5$, but the right side is $2 + 2 = 4$.
- **Sequences in closed intervals**: if $x_n \geq a$ for all n and $x_n \to p$, then $p \geq a$. If instead $p < a$, take $\varepsilon = a - p$; some x_n has $|x_n - p| < \varepsilon$, giving $x_n < a$, a contradiction.

Section 3: Metric space fundamentals

How to build metrics, how to compare spaces (isometries, Lipschitz maps), and boundedness, balls and sequences in an *abstract* metric space.

Our stock of examples

Space	Metric	Induced by a norm?
$\mathbb{R}$	$\lvert y-x\rvert$ (standard)	yes, $\lvert\cdot\rvert$
$\mathbb{R}^N$, $\ell^p(N)$, $1 \le p < \infty$	$d_p(x,y) = \left(\sum \lvert y_i - x_i\rvert^p\right)^{1/p}$; the Triangle Inequality is **Minkowski's inequality**	yes, $\lVert\cdot\rVert_p$
$\mathbb{R}^N$, $\ell^\infty(N)$	$d_\infty(x,y) = \max \lvert y_i - x_i\rvert$	yes, $\lVert\cdot\rVert_\infty$
$C(I;\mathbb{R})$	$d_1(f,g) = \int \lvert g-f\rvert$	yes, L^1-norm
$C(I;\mathbb{R})$	$d_{\sup}$ (uniform distance)	yes, sup-norm
any set	discrete	no (not a vector space in general)
$\mathcal{K}(\mathbb{R}^2)$	Hausdorff	no
$\mathbb{R}$	$\lvert \tan^{-1} y - \tan^{-1} x\rvert$ (bounded, $< \pi$)	**no**

The case $p = 2$ is the Euclidean metric.

New metrics from old

METRIC INDUCED BY A NORM (EXERCISE 3.1)

$d(x,y) = \lVert y - x\rVert$ is a metric. Later in the course: every metric space is isometric to a subset of a real normed space.

PROPOSITION (METRIC INDUCED BY AN INJECTION)

If $\phi : A \to E$ is **injective** into a metric space, then $d(s,t) := d_E(\phi(s), \phi(t))$ is a metric on A.

Examples: the standard metric on $\mathbb{C}$, via $\mathbb{C} \to \mathbb{R}^2$, $x_1 + ix_2 \mapsto (x_1, x_2)$, which recovers $\lvert v-u\rvert$. The bounded metric $\lvert \tan^{-1} y - \tan^{-1} x\rvert$ on $\mathbb{R}$, via the injection $\tan^{-1} : \mathbb{R} \to \mathbb{R}$.

PROPOSITION (NEW METRIC VIA AN INCREASING SUBADDITIVE FUNCTION)

Let $\alpha : [0, \infty[\to [0, \infty[$ satisfy

- $\alpha(s) = 0 \iff s = 0$;
- **increasing**: $s \leq t \Rightarrow \alpha(s) \leq \alpha(t)$;
- **subadditive**: $\alpha(s + t) \leq \alpha(s) + \alpha(t)$.

Then $d' = \alpha \circ d$ is a metric for **every** metric d.

Proof. Symmetry is immediate. (MS1): $\alpha(d(s,t)) = 0 \iff d(s,t) = 0 \iff s = t$. (MS3): $\alpha(d(r,t)) \leq \alpha(d(r,s) + d(s,t)) \leq \alpha(d(r,s)) + \alpha(d(s,t))$, using **increasing** then **subadditive**. ∎

Examples: at ($a > 0$); $\min\{t, 1\}$; $t/(1+t)$; $\tan^{-1} t$; $t^{1/p}$ ($p \geq 1$); and 1 for $t \neq 0$, 0 for $t = 0$ (this always gives the discrete metric). The 2nd, 3rd and 4th give **bounded metrics**. In the 3rd, 4th and 5th, α is invertible.

METRIC SUBSPACE

For $A \subset S$, $d|_{A \times A}$ is the **metric induced on** A, and $(A, d|_{A \times A})$ is a **metric subspace**.

Warning: a subspace can look completely different from the whole space. The standard basis vectors $\{e_1, \ldots, e_n\} \subset \mathbb{R}^n$ get the metric $\sqrt{2}\, d_{\text{discr}}$. This is unlike linear subspaces.

PRODUCT METRICS

Finite products: $d_p(x, y) = \|(d_1(x_1, y_1), \ldots, d_N(x_N, y_N))\|_p$, for $p \in [1, \infty]$.

Countable products: $d(x, y) = \sum_{k \geq 1} 2^{-k} \frac{d_k(x_k, y_k)}{1 + d_k(x_k, y_k)}$ (Exercise 3.5).

Isometries, Lipschitz maps, contractions

DEFINITIONS

Isometry: $d_F(f(x), f(y)) = d_E(x, y)$ for all x, y. E and F are **isometric** if there is an isometric bijection. Isometric spaces are indistinguishable as metric spaces (like isomorphic vector spaces). Examples: inclusions of subspaces; rotations, reflections and translations of $\mathbb{E}^N$; $f \mapsto f(-\cdot)$ on $(C[-1, 1], d_{\sup})$.

Lipschitz: $\exists M\ \forall x, y :\ d_F(f(x), f(y)) \leq M\, d_E(x, y)$. The **Lipschitz constant** is $\operatorname{Lip} f = \sup\{d_F(f(x), f(y))/d_E(x, y) : x \neq y\}$.

Strict contraction: Lipschitz with $\operatorname{Lip} f < 1$.

Bi-Lipschitz ("Lipeomorphism"): a Lipschitz bijection whose inverse is Lipschitz.

Isometries are Lipschitz (constant 1), so isometric bijections are bi-Lipschitz. $\operatorname{Lip} f = 0$ means f is **constant**.

EXAMPLE (CONTINUOUSLY DIFFERENTIABLE ⇒ LIPSCHITZ)

If $f \in C^1(I;\mathbb{R})$, then $|f'| \le M$ (Uniform Continuity Theorem). By FTC, $|f(t)-f(s)| = |\int_s^t f'| \le M|t-s|$, so $\operatorname{Lip} f \le d_{\sup}(f',0)$. Hence $C^1(I;\mathbb{R}) \subset C_{\mathrm{Lip}}(I;\mathbb{R}) \subset C(I;\mathbb{R}) \subset \mathcal{F}_b(I;\mathbb{R})$.

EXAMPLE (LIPSCHITZ BIJECTION THAT IS NOT BI-LIPSCHITZ)

$f : I \to I$, $x \mapsto x^2$. It is Lipschitz since $\frac{|y^2-x^2|}{|y-x|} = |x+y| \le 2$. Its inverse $g(t) = \sqrt{t}$ is not Lipschitz: by the Mean Value Theorem, for $s,t \in]0,1/M[$ we get $\frac{|g(t)-g(s)|}{|t-s|} = \frac{1}{2\sqrt{u}} \ge \frac{\sqrt{M}}{2}$ for some u between them. M is arbitrary, so $\operatorname{Lip} g = \infty$.

DISTANCE FROM A SET

For nonempty A: $\operatorname{dist}_A(x) = \inf\{d(x,a) : a \in A\}$. It is 1-Lipschitz (Exercise 3.7). $\operatorname{dist}_A(x) = 0$ iff some sequence in A converges to x. Later this is the statement "$x \in \overline{A}$".

Boundedness and the paradigm space $\ell^\infty(S;E)$

DIAMETER, BOUNDED

$\operatorname{diam} A = \sup\{d(x,y) : x,y \in A\}$ (possibly ∞). A is **bounded** if $\operatorname{diam} A < \infty$. A metric (space) is bounded if E itself is.

Facts:

- Discrete spaces are bounded.
- The supremum need not be attained.
- A is bounded iff $\sup_{a\in A} d(p,a) < \infty$ for some (equivalently, all) p.
- $A \subset B \Rightarrow \operatorname{diam} A \le \operatorname{diam} B$.

PROPOSITION (THE METRIC SPACE $\ell^\infty(S;E)$)

$\mathcal{F}_b(S;E)$ is the set of bounded functions $S \to E$ (those with $\operatorname{diam}\operatorname{Ran} f < \infty$). The sup metric $d_{\sup}(f,g) = \sup_{s\in S} d_E(f(s),g(s))$ is a metric on it.

Proof.

Finite by Exercise 3.14.

(MS1): $d_{\sup} = 0 \iff \forall s:\ d_E(f(s),g(s)) = 0 \iff f = g$.

(MS2): pointwise symmetry.

(MS3): for each t, $d_E(f(t),h(t)) \le d_E(f(t),g(t)) + d_E(g(t),h(t)) \le d_{\sup}(f,g) + d_{\sup}(g,h)$. The right side is independent of t; take the sup over t. ∎

$\ell^\infty(\{1..N\};\mathbb{R}) \cong \ell^\infty(N)$ isometrically, so this massively extends the max metric. Later in the course: every metric space embeds isometrically in some $\ell^\infty(S;\mathbb{R})$.

Balls and sequences

BALLS AND SPHERES

Open ball $B_r(a) = \{x : d(x,a) < r\}$. **Closed ball** $B_r[a] = \{x : d(x,a) \le r\}$. **Sphere** $S_r(a) = \{x : d(x,a) = r\}$.

The open ball and the sphere partition the closed ball. For $R > r$: $\{a\} \subset B_r(a) \subset B_r[a] \subset B_R(a)$.

Balls do not usually look like balls: in a discrete space, $B_r(a) = \{a\}$ if $r \le 1$ and $B_r(a) = E$ if $r > 1$.

DIAMETERS OF BALLS

$\operatorname{diam} B_r[a] \le \operatorname{diam} B_R(a) \le \operatorname{diam} B_R[a] \le 2R$ for $R > r$.

In a discrete space, $\operatorname{diam} B_r(a)$ is 0 if $r \le 1$ or $E = \{a\}$, and 1 if $r > 1$ and $\#E \ge 2$. So diam can be much less than $2r$.

In normed spaces, $\operatorname{diam} B_R(a) = \operatorname{diam} B_R[a] = 2R$ (no surprises).

OPENNESS PROPERTY OF OPEN BALLS

$\forall p \in B\ \exists r > 0:\ B_r(p) \subset B$ (Exercise 3.10 with $b = a$, $s = r$).

SEQUENCES IN A METRIC SPACE

Bounded: the set of terms is bounded, equivalently $(d(a, x_n))_n$ is bounded in $\mathbb{R}$ for some or all a.

Cauchy: $\forall \varepsilon > 0\ \exists N\ \forall n, m \geq N:\ d(x_n, x_m) < \varepsilon$. Equivalently, $\lim_N \sup_p d(x_N, x_{N+p}) = 0$.

Convergent to a: $d(x_n, a) \to 0$ in $\mathbb{R}$.

Proposition: convergent ⇒ Cauchy ⇒ bounded (Exercise 3.11). The converses fail in general metric spaces.

Exercises for Section 3, with full solutions

EXERCISE 3.1 (METRIC INDUCED BY A NORM)

5 marks

▾ Solution

$d(x, y) = \|y - x\| \in [0, \infty[$.

(MS1): $\|y - x\| = 0 \iff y - x = 0 \iff x = y$, by (N1) and $\|0\| = 0$.

(MS2): $\|x - y\| = \|(-1)(y - x)\| = |-1|\,\|y - x\|$, by (N2).

(MS3): $\|z - x\| = \|(z - y) + (y - x)\| \leq \|z - y\| + \|y - x\|$, by (N3). ∎

EXERCISE 3.2 (METRIC INDUCED BY AN INJECTION)

▾ Solution

$d(s, t) = d_E(\phi(s), \phi(t)) \geq 0$ and is finite.

(MS1): $d(s, t) = 0 \iff \phi(s) = \phi(t) \iff s = t$. The last step is exactly **injectivity**; without it, (MS1b) fails.

(MS2) and (MS3) are inherited directly from d_E, applied to the points $\phi(r), \phi(s), \phi(t)$. ∎

EXERCISE 3.3 (ASSOCIATED BOUNDED METRIC, $A(T) = T/(1+T)$)

10 marks

▾ Solution

Check the three hypotheses of the Proposition.

$\alpha(t) = 0 \iff t = 0$.

Increasing: $\alpha(t) = 1 - \frac{1}{1+t}$ and $\frac{1}{1+t}$ decreases.

Subadditive: $\alpha(s+t) = \frac{s}{1+s+t} + \frac{t}{1+s+t} \le \frac{s}{1+s} + \frac{t}{1+t}$, since enlarging a positive denominator decreases the fraction.

Hence $\alpha \circ d$ is a metric, and it is bounded by 1. ∎

EXERCISE 3.4 (REARRANGED TRIANGLE INEQUALITY, ℓ^1, ℓ^∞, CONVERGENCE)

15 marks

▾ Solution

(a) **Rearranged Triangle Inequality**: $|d(x,z) - d(y,z)| \le d(x,y)$. By (MS3), $d(x,z) \le d(x,y) + d(y,z)$, so $d(x,z) - d(y,z) \le d(x,y)$. Swapping x and y and using (MS2): $d(y,z) - d(x,z) \le d(x,y)$. Combine the two. For norms: $\big|\|x\| - \|y\|\big| \le \|x - y\|$ (take $z = 0$).

(b)(i) $d_1(x,y) = \sum |y_i - x_i|$. (MS1): a sum of nonnegative terms is zero iff each term is. (MS2): $|x_i - y_i| = |y_i - x_i|$. (MS3): add the inequalities $|z_i - x_i| \le |y_i - x_i| + |z_i - y_i|$ over i.

(b)(ii) $d_\infty = \max_i |y_i - x_i|$. (MS1): the max is zero iff all terms are. (MS2) is clear. (MS3): for each i, $|z_i - x_i| \le |y_i - x_i| + |z_i - y_i| \le d_\infty(x,y) + d_\infty(y,z)$; take the max over i.

(c) "$x_n \to a$" means $d(x_n, a) \to 0$ in $\mathbb{R}$, i.e. $\forall \varepsilon > 0\ \exists N\ \forall n \ge N :\ d(x_n, a) < \varepsilon$. ∎

EXERCISE 3.5 (METRIC ON A COUNTABLE PRODUCT)

▾ Solution

$d(x,y) = \sum_k (\alpha_k \circ d_k)(x_k, y_k)$ with $\alpha_k(t) = 2^{-k}\frac{t}{1+t}$. Each α_k satisfies the hypotheses of the Proposition (a positive multiple of the function in Exercise 3.3), so each $\rho_k := \alpha_k \circ d_k$ is a metric on S_k with $\rho_k \le 2^{-k}$. The series therefore converges by comparison with $\sum 2^{-k} = 1$, and $0 \le d \le 1$.

(MS1): $d(x,y) = 0 \iff$ every term is $0 \iff \forall k:\ x_k = y_k \iff x = y$ (equality in a product is componentwise).

(MS2): termwise.

(MS3): for each K, $\sum_{k\le K} \rho_k(x_k,z_k) \le \sum_{k\le K} \rho_k(x_k,y_k) + \sum_{k\le K} \rho_k(y_k,z_k) \le d(x,y) + d(y,z)$. Let $K \to \infty$ (order preservation under limits). ∎

EXERCISE 3.6 (ISOMETRIC BIJECTIONS HAVE ISOMETRIC INVERSES)

▾ Solution

(a) $f(x) = f(y) \Rightarrow d_E(x,y) = d_F(f(x),f(y)) = 0 \Rightarrow x = y$. So **every isometry is injective**.

(b) For $u, v \in F$: $d_E(g(u),g(v)) = d_F(f(g(u)),f(g(v))) = d_F(u,v)$, using the isometry property and then $f \circ g = \mathrm{id}_F$. ∎

EXERCISE 3.7 ("DISTANCE FROM A SET" FUNCTION)

▾ Solution

(a) For any $a \in A$: $\mathrm{dist}_A(x) \le d(x,a) \le d(x,y) + d(y,a)$. So $\mathrm{dist}_A(x) - d(x,y)$ is a lower bound for $\{d(y,a) : a \in A\}$, giving $\mathrm{dist}_A(x) - d(x,y) \le \mathrm{dist}_A(y)$. By symmetry, $|\mathrm{dist}_A(x) - \mathrm{dist}_A(y)| \le d(x,y)$. Hence dist_A is Lipschitz with $\mathrm{Lip} \le 1$.

(b) (i)⇒(ii): by the infimum version of Approximation of suprema by sequences, there are $a_n \in A$ with $d(x,a_n) \to \mathrm{dist}_A(x) = 0$. Concretely, pick $d(x,a_n) < 1/n$. (ii)⇒(i): $0 \le \mathrm{dist}_A(x) \le d(x,a_n) \to 0$, so $\mathrm{dist}_A(x) = 0$ by order preservation.

(c) Apply (b) at every $x \in E$. ∎ (Later: (ii) says A is **dense** in E.)

EXERCISE 3.8 (DIAMETER OF A UNION)

▾ Solution

(a) $A, B \subset A \cup B$, and diam is monotone under inclusion.

(b) Assume $A, B \neq \emptyset$ and both are bounded (otherwise the inequality is trivial). Let $x, y \in A \cup B$. If both are in A, $d(x,y) \leq \operatorname{diam} A$, which is at most the right side; similarly if both are in B. If $x \in A$ and $y \in B$, then for any $a \in A$, $b \in B$: $d(x,y) \leq d(x,a) + d(a,b) + d(b,y) \leq \operatorname{diam} A + d(a,b) + \operatorname{diam} B$. Taking the infimum over a, b gives $d(x,y) \leq \operatorname{diam} A + I + \operatorname{diam} B$. Take the sup over x, y. ∎

EXERCISE 3.9 (UNIT SPHERES IN $\ell^P(2)$)

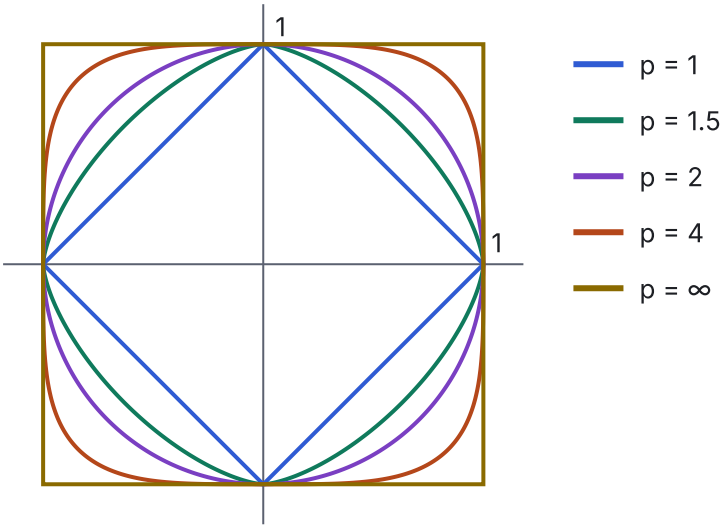
10 marks

▾ Solution (picture)

$S_1(0) = \{x : |x_1|^p + |x_2|^p = 1\}$ (or $\max\{|x_1|, |x_2|\} = 1$ for $p = \infty$).

- $p = 1$: a **diamond** (square rotated 45°) with vertices $(\pm 1, 0), (0, \pm 1)$.
- $p = 2$: the **circle**.
- $p = \infty$: the **square** with corners $(\pm 1, \pm 1)$.
- $1 < p < 2$: a convex curve between the diamond and the circle.
- $2 < p < \infty$: a convex curve between the circle and the square.

As p increases, the spheres grow outward: since $\|x\|_\infty \leq \|x\|_q \leq \|x\|_p$ for $p \leq q$, the ball $B_1^{(p)} \subset B_1^{(q)}$. All of them pass through $(\pm 1, 0)$ and $(0, \pm 1)$.



EXERCISE 3.10 (BALLS IN AN INTERSECTION OF BALLS)

▾ Solution

Take $t = \min\{r - d(p,a),\ s - d(p,b)\}$. It is > 0 because p is in both open balls. If $d(x,p) < t$, then $d(x,a) \le d(x,p) + d(p,a) < t + d(p,a) \le r$, and similarly $d(x,b) < s$. So $B_t(p) \subset B_r(a) \cap B_s(b)$. ∎

EXERCISE 3.11 (CONVERGENT ⇒ CAUCHY ⇒ BOUNDED)

▾ Solution

(a) If $x_n \to a$ and $\varepsilon > 0$, choose N with $d(x_n, a) < \varepsilon/2$ for $n \ge N$. Then $d(x_n, x_m) \le d(x_n, a) + d(a, x_m) < \varepsilon$.

(b) With $\varepsilon = 1$, choose N with $d(x_n, x_m) < 1$ for $n, m \ge N$. Let $R = \max\{d(x_1, x_N), \dots, d(x_{N-1}, x_N), 1\}$. Then every $x_n \in B_R[x_N]$, so $\operatorname{diam}\{x_n\} \le 2R < \infty$. ∎

EXERCISE 3.12 (CONVERGENCE OF METRIC VALUES)

▾ Solution

(a) By the Rearranged Triangle Inequality, $|d(x_n,p) - d(x_m,p)| \le d(x_n, x_m)$. So $(d(x_n,p))$ is **Cauchy in $\mathbb{R}$**, hence convergent by the **completeness of $\mathbb{R}$**.

(b) $|d(x_n,p) - d(a,p)| \le d(x_n, a) \to 0$, so $d(x_n,p) \to d(a,p)$ by the Sandwich Principle. ∎

EXERCISE 3.13 (CONVERGENCE IN PRODUCT SPACES)

▾ Solution

The sum metric is $d(x,y) = d_1(x_1,y_1) + d_2(x_2,y_2)$.

(i)⇒(ii): if $x^{(n)} \to a = (a_1, a_2)$, then $0 \le d_i(x_i^{(n)}, a_i) \le d(x^{(n)}, a) \to 0$. Sandwich.

(ii)⇒(i): if $x_i^{(n)} \to a_i$, then $d(x^{(n)}, (a_1,a_2)) = d_1(\cdot) + d_2(\cdot) \to 0 + 0$, by the algebra of limits. ∎

EXERCISE 3.14 (FINITENESS OF THE SUP METRIC ON $\mathcal{F}_B(S;E)$)

5 marks

▾ Solution

If $S = \emptyset$ there is nothing to prove (the sup of the empty set of nonnegative numbers is taken to be 0). Otherwise fix $s_0 \in S$. For every s: $d(f(s), g(s)) \le d(f(s), f(s_0)) + d(f(s_0), g(s_0)) + d(g(s_0), g(s)) \le \operatorname{diam} \operatorname{Ran} f + d(f(s_0), g(s_0)) + \operatorname{diam} \operatorname{Ran} g$. This is a finite constant independent of s, so the sup is finite. ∎

IN-TEXT QUESTIONS

▾ Answers

- **Lipschitz constant 0?** Then $d(f(x), f(y)) \le 0$, so $f(x) = f(y)$ for all x, y: f is constant.
- **Basis vectors in $\mathbb{R}^n$:** $\|e_i - e_j\|_2 = \sqrt{1+1} = \sqrt{2}$ for $i \neq j$, so $d' = \sqrt{2}\, d_{\text{discr}}$.
- **Why is the $\tan^{-1}$ metric not induced by a norm?** It is bounded (all distances are $< \pi$). A norm-metric on a nonzero space is unbounded, since $d(0, \lambda v) = |\lambda| \|v\| \to \infty$.

Section 4: Metric space topology

Open and closed sets, interior, closure, boundary, continuity through open sets, density, separability, homeomorphisms, equivalence of metrics, product spaces and subspaces. This is the most examinable section.

Open sets and interior

DEFINITION (OPEN SET)

$U \subset E$ is **open** (in E) if $\forall p \in U\ \exists r > 0 :\ B_r(p) \subset U$. In words: **wherever you are in U, there is a ball centred at your location and contained entirely in U.** $\mathcal{O}(E)$ denotes the collection of open sets.

EXAMPLE (OPEN BALLS ARE OPEN SETS)

Proof. Let $p \in B_R(a)$ and $r = R - d(p,a) > 0$. If $d(x,p) < r$, then $d(x,a) \le d(x,p) + d(p,a) < r + d(p,a) = R$. So $B_r(p) \subset B_R(a)$. ∎

In $\mathbb{R}$ (standard metric), the open balls are exactly the bounded open intervals: $B_r(x) =]x-r, x+r[$, and $]c,d[= B_{(d-c)/2}(\frac{c+d}{2})$.

PROPOSITION (THE FAMILY OF OPEN SETS)

(a) $\emptyset, E \in \mathcal{O}$; (b) **arbitrary unions** of open sets are open; (c) the intersection of **two** (hence, by induction, finitely many) open sets is open.

Proof. (a) Immediate: for $\emptyset$ the condition is vacuous; for E every ball lies in E. (b) If $p \in \bigcup \mathcal{U}$, then $p \in U$ for some $U \in \mathcal{U}$, and $B_r(p) \subset U \subset \bigcup \mathcal{U}$. (c) If $p \in U \cap V$ with $B_r(p) \subset U$ and $B_s(p) \subset V$, take $t = \min\{r, s\}$. ∎

PITFALL

Infinite intersections of open sets need not be open: $\bigcap_n\,]-\frac{1}{n}, \frac{1}{n}[= \{0\}$ is not open in $\mathbb{R}$. The proof of (c) breaks because $\inf_n r_n$ may be 0.

PROPOSITION

Every open set is a union of open balls: $U = \bigcup_{x \in U} B_{r(x)}(x)$.

INTERIOR

p is an **interior point** of A if $\exists r > 0 :\ B_r(p) \subset A$. The **interior** $\operatorname{Int} A$ is the set of interior points, and $\operatorname{Int} A \subset A$.

Example: $\operatorname{Int} \mathbb{Q} = \emptyset$ in $\mathbb{R}$, because every interval $]x - r, x + r[$ contains irrationals (density of $\mathbb{R} \setminus \mathbb{Q}$).

PROPOSITION (INTERIOR = LARGEST OPEN SUBSET)

$\operatorname{Int} A = \bigcup \{U \subset A : U \text{ open}\}$. Hence (a) $\operatorname{Int} A$ is open; (b) $A = \operatorname{Int} A \iff A$ is open; (c) $\operatorname{Int}(\operatorname{Int} A) = \operatorname{Int} A$.

Proof. If $p \in \operatorname{Int} A$, then $B_r(p)$ is an open subset of A containing p. Conversely, if $p \in U \subset A$ with U open, then $B_r(p) \subset U \subset A$. ∎

Continuity

CONTINUITY BETWEEN METRIC SPACES

$f : E \to E'$ is **continuous at** p if $\forall \varepsilon > 0\ \exists \delta > 0 :\ f(B_\delta(p)) \subset B_\varepsilon(f(p))$. It is **sequentially continuous at** p if $x_n \to p \Rightarrow f(x_n) \to f(p)$. **Continuous** means continuous at every p.

PROPOSITION: CONTINUOUS AT P ⟺ SEQUENTIALLY CONTINUOUS AT P

Proof. (⇒) Given ε, take δ, then N with $d(x_n, p) < \delta$ for $n \geq N$. Then $f(x_n) \in B_\varepsilon(f(p))$. (⇐, contrapositive) If f is not continuous at p, there is $\eta > 0$ such that for every n some $x_n \in B_{1/n}(p)$ has $d(f(x_n), f(p)) \geq \eta$. Then $x_n \to p$ but $f(x_n) \not\to f(p)$. ∎

PROPOSITION (LIPSCHITZ ⇒ CONTINUOUS)

$0 \leq d(f(x_n), f(a)) \leq L\, d(x_n, a) \to 0$. Alternatively, $f(B_r(p)) \subset B_{Lr}(f(p))$, so take $\delta = \varepsilon / L$ (any δ works if $L = 0$). So $C_{\text{Lip}}(E; E') \subset C(E; E')$.

THEOREM (GLOBAL CONTINUITY VIA OPEN SETS). KEY.

f is continuous $\iff$ $f^{-1}(V)$ is open in E for every open V in E'.

Proof.

($\Rightarrow$) Let $p \in f^{-1}(V)$. Since V is open, $B_\varepsilon(f(p)) \subset V$ for some ε. Continuity gives δ with $f(B_\delta(p)) \subset B_\varepsilon(f(p))$. So $B_\delta(p) \subset f^{-1}(f(B_\delta(p))) \subset f^{-1}(B_\varepsilon(f(p))) \subset f^{-1}(V)$.

($\Leftarrow$) $f^{-1}(B_\varepsilon(f(p)))$ is open (balls are open) and contains p, so it contains some $B_\delta(p)$. Then $f(B_\delta(p)) \subset f(f^{-1}(B_\varepsilon(f(p)))) \subset B_\varepsilon(f(p))$. ∎

Uses: $f^{-1}(f(A)) \supset A$ and $f(f^{-1}(B)) \subset B$ (Exercise 4.3).

PITFALL

It is **inverse** images of open sets that are open. **Direct** images of open sets under continuous maps need not be open: a constant map $\mathbb{R} \to \mathbb{R}$ sends $\mathbb{R}$ to a single point, which is not open.

Closed sets, closure, boundary

PROPOSITION (POINT OF CLOSURE)

TFAE for $A \subset E$, $p \in E$: (i) $\forall r > 0 :\ B_r(p) \cap A \neq \emptyset$; (ii) some sequence (a_n) in A has $a_n \to p$; (iii) $\operatorname{dist}_A(p) = 0$.

Proof. (iii)⇒(ii): approximation of the infimum by a sequence. (ii)⇒(i): $a_N \in B_r(p) \cap A$ for large N. (i)⇒(iii): for every ε, some $a \in A$ has $d(a,p) < \varepsilon$, so $\operatorname{dist}_A(p) < \varepsilon$. ∎

CLOSURE, CLOSED

p is a **point of closure** of A if (i) holds. The **closure** is $\overline{A} = \{\text{points of closure of } A\}$, and $A \subset \overline{A}$.

A is **closed** if $\overline{A} = A$. Equivalently: **whenever a sequence in A converges in E, its limit lies in A.** This is important to grasp and remember. $\mathcal{C}(E)$ denotes the closed sets.

FACTS ABOUT CLOSURES

- $\overline{\overline{A}} = \overline{A}$, so **closures are closed**. Proof: if $p \in \overline{\overline{A}}$, pick $v \in \overline{A}$ with $d(v,p) < r/2$, then $a \in A$ with $d(a,v) < r/2$; so $d(a,p) < r$.
- $\overline{B_r(a)} \subset B_r[a]$. Proof: if $x_n \to p$ with $d(x_n,a) < r$, then $d(p,a) = \lim d(x_n,a) \leq r$ (convergence of metric values, order preservation).
- In general $B_r(a) \subset \overline{B_r(a)} \subset B_r[a]$, and both inclusions can be strict. Discrete space: $\overline{B_1(a)} = B_1(a) = \{a\}$, but $B_1[a] = E$.
- In a **normed** space, $\overline{B_r(a)} = B_r[a]$.

THEOREM (CLOSED = COMPLEMENT OF OPEN)

A is closed $\iff A^{\complement} = E \setminus A$ is open.

Proof. ($\Rightarrow$) If $p \notin A = \overline{A}$, then $r := \operatorname{dist}_A(p) > 0$, so $B_r(p) \subset A^{\complement}$. ($\Leftarrow$, contrapositive) If A is not closed, pick $p \in \overline{A} \setminus A$. Then $p \in A^{\complement}$, but every ball around p meets A. So $A^{\complement}$ is not open. ∎

MOST SETS IN A GENERAL METRIC SPACE ARE NEITHER OPEN NOR CLOSED

Examples: $[c,d[$ in $\mathbb{R}$, and $\mathbb{Q}$ in $\mathbb{R}$. "Not open" does **not** mean "closed". Some sets are both ($\emptyset$, E, and every subset of a discrete space).

COROLLARY (THE FAMILY OF CLOSED SETS)

(a) $\emptyset$, E are closed; (b) **arbitrary intersections** of closed sets are closed; (c) **finite unions** of closed sets are closed. Proof: de Morgan applied to the open-set properties.

Infinite unions can fail: $\bigcup_n [\frac{1}{n}, 1] =]0,1]$ is not closed.

PROPOSITION (CLOSURE = SMALLEST CLOSED SUPERSET)

$\overline{A} = \bigcap \{F \supset A : F \text{ closed}\}$.

Proof. Let H be the right side. $\overline{A}$ is a closed set containing A, so $H \subset \overline{A}$. Conversely, if $p = \lim a_n$ with $a_n \in A \subset H$, then since H is closed, $p \in H$. ∎

Corollary: $(\overline{A})^{\complement} = \operatorname{Int}(A^{\complement})$ and $(\operatorname{Int} A)^{\complement} = \overline{A^{\complement}}$.

TOPOLOGICAL BOUNDARY

$\partial A := \overline{A} \setminus \operatorname{Int} A = \overline{A} \cap \overline{A^{\complement}}$. It is **closed** (an intersection of two closed sets). $\operatorname{Int} A$, ∂A and $\operatorname{Int} A^{\complement}$ form a **partition of** E (allowing empty pieces).

Examples in $\mathbb{R}$: $\partial[0,1[= \{0,1\}$; $\partial\mathbb{Q} = \mathbb{R}$ (closure $\mathbb{R}$, interior $\emptyset$).

Density and separability

DENSE, SEPARABLE

For $D \subset A \subset E$: D is **dense in** A if $\overline{D} \supset A$. **Test**: TFAE (i) D is dense in A; (ii) $\forall a \in A\, \forall R > 0$: $B_R(a) \cap D \neq \emptyset$; (iii) every $a \in A$ is a limit of a sequence in D.

E is **separable** if it has a **countable dense subset**.

Example: $\mathbb{Q} \cap I$ and $(\mathbb{R} \setminus \mathbb{Q}) \cap I$ are both dense in I, yet they are disjoint.

PROPOSITION (DENSE ∩ OPEN-DENSE = DENSE)

If U is open and dense and A is dense, then $U \cap A$ is dense.

Proof. Let $p \in E$ and $R > 0$. (1) Density of U gives $u \in B_R(p) \cap U$. (2) $B_R(p) \cap U$ is open, so some $B_r(u) \subset B_R(p) \cap U$. (3) Density of A gives $a \in B_r(u) \cap A \subset B_R(p) \cap U \cap A$. ∎ (Openness of U is essential: $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are both dense, but their intersection is empty.)

EXAMPLES OF SEPARABLE SPACES

(1) $\ell^p(N)$ for every $p \in [1, \infty]$: $\mathbb{Q}^N$ is countable and dense. (2) $(C(I;\mathbb{R}), d_{\sup})$: polynomials are dense by the **Weierstrass Approximation Theorem**, and polynomials with rational coefficients approximate them uniformly. A second proof uses piecewise linear functions (Exercise 4.11).

Open subsets of $\mathbb{R}$

THEOREM (IN $\mathbb{R}$, OPEN ≡ COUNTABLE UNION OF OPEN INTERVALS)

Let $\mathcal{Q} = \{]q, r[: q, r \in \mathbb{Q}, q < r\}$, which is countable (it injects into $\mathbb{Q}^2$). For any $A \subset \mathbb{R}$, let $\mathcal{Q}_A = \{J \in \mathcal{Q} : J \subset A\}$. Then $\bigcup \mathcal{Q}_A = \operatorname{Int} A$.

Proof. ($\subset$) Each J in $\mathcal{Q}_A$ is an open subset of A. ($\supset$) If $]x - R, x + R[\subset A$, density of $\mathbb{Q}$ gives rationals $x - R < q < x < r < x + R$, so $x \in]q, r[\in \mathcal{Q}_A$. For open U, $U = \operatorname{Int} U = \bigcup \mathcal{Q}_U$, a countable union of open intervals. ∎

Challenge ($\mathbb{R}^N$): every open set in $\mathbb{R}^N$ is a countable union of open balls with rational centres and rational radii (or of open boxes with rational corners). The same proof works using density of $\mathbb{Q}^N$.

Homeomorphisms and equivalence of metrics

HOMEOMORPHISM

$f : E \to E'$ is a **homeomorphism** if (H1) f is continuous, (H2) f is bijective, (H3) f^{-1} is continuous.

- (H3) does **not** follow from (H1) and (H2). Example: $\mathrm{id} : (\mathbb{R}, d_{\text{discrete}}) \to (\mathbb{R}, |\cdot|)$ is a continuous bijection (every map out of a discrete space is continuous), but its inverse is not continuous ($\{0\}$ is open in the discrete space but not in $\mathbb{R}$).
- Bi-Lipschitz maps are homeomorphisms.
- $\tan :]-\frac{\pi}{2}, \frac{\pi}{2}[\to \mathbb{R}$ is a homeomorphism whose inverse is Lipschitz.
- For a continuous bijection, TFAE: it is a homeomorphism; it maps open sets to open sets; it maps closed sets to closed sets.

TOPOLOGICAL VS LIPSCHITZ EQUIVALENCE OF METRICS

Let d_1, d_2 be metrics on S.

(a) They are **topologically equivalent** if $\mathrm{id}_S : (S, d_1) \to (S, d_2)$ is a **homeomorphism**.

(b) They are **Lipschitz equivalent** if id_S is **bi-Lipschitz**, i.e. $\exists L, K > 0$ with $d_2 \le L d_1$ and $d_1 \le K d_2$.

Lipschitz equivalent ⇒ topologically equivalent (the converse is false). Equivalent metrics have the **same open sets**.

PROPOSITION (EQUIVALENCE VIA CONVERGENCE)

TFAE: (i) d_1, d_2 are topologically equivalent; (ii) for all p and (x_n): $x_n \xrightarrow{d_1} p \iff x_n \xrightarrow{d_2} p$; (iii) d_1-closed $\iff$ d_2-closed. (Also (iv): d_1-open $\iff$ d_2-open, Exercise 4.7.)

PROPOSITION (LIPSCHITZ EQUIVALENCE PRESERVES BOUNDEDNESS AND CAUCHY)

If $d_2 \leq Ld_1$, then $d_2\text{-}\operatorname{diam} A \leq L\, d_1\text{-}\operatorname{diam} A$ and $d_2(x_n, x_m) \leq Ld_1(x_n, x_m)$. Apply this both ways.

TOPOLOGICAL EQUIVALENCE DOES NOT PRESERVE BOUNDEDNESS OR CAUCHYNESS

Boundedness: d and $\min\{d, 1\}$ are topologically equivalent on $\mathbb{R}$, but $\mathbb{R}$ is unbounded for d and bounded for $\min\{d, 1\}$.

Cauchy: $d_{NS}(x, y) = |\tan^{-1} y - \tan^{-1} x|$ is topologically equivalent to the standard metric on $\mathbb{R}$. But $x_n = n$ is d_{NS}-Cauchy (since $\tan^{-1} n \to \pi/2$ converges) and not Cauchy in the standard metric. So the two metrics are not Lipschitz equivalent.

An **unbounded** metric cannot have a Lipschitz equivalent **bounded** metric: if $d \leq Kd'$ and $d' \leq M$, then $d \leq KM$.

EXAMPLES YOU MUST KNOW

All ℓ^p metrics on $\mathbb{R}^N$ are Lipschitz equivalent, since $\|x\|_\infty \leq \|x\|_p \leq N^{1/p}\|x\|_\infty$. (Proof: $|x_j|^p \leq \sum |x_i|^p \leq N \max |x_i|^p$.)

Corollaries: for $A \subset \mathbb{R}^N$, $p, q \in [1, \infty]$: ℓ^p-open $\iff$ ℓ^q-open; ℓ^p-closed $\iff$ ℓ^q-closed; ℓ^p-bounded $\iff$ ℓ^q-bounded. A sequence is ℓ^p-convergent to a iff it converges **coordinatewise** ($x_i^{(n)} \to a_i$ for each i). The same holds for Cauchy, and bounded means $\exists L :\ |x_i^{(n)}| \leq L$ for all i, n.

▾ Proof of the coordinatewise criterion (likely exam question)

For every i: $|x_i^{(n)} - a_i| \leq \|x^{(n)} - a\|_\infty \leq \|x^{(n)} - a\|_p \leq N^{1/p}\|x^{(n)} - a\|_\infty \leq N^{1/p} \sum_j |x_j^{(n)} - a_j|$.

If $\|x^{(n)} - a\|_p \to 0$, each coordinate converges (Sandwich). If every coordinate converges, the finite sum $\sum_j |x_j^{(n)} - a_j| \to 0$ (algebra of limits), so $\|x^{(n)} - a\|_p \to 0$.

The Cauchy and bounded statements follow in the same way, replacing a by $x^{(m)}$ or by 0. ∎

PROPOSITION (CONDITIONS FOR A∘D TO BE AN EQUIVALENT METRIC)

Suppose $\alpha : [0,\infty[\to [0,\infty[$ is (i) definite ($\alpha(s) = 0 \iff s = 0$), (ii) increasing, (iii) subadditive, and (iv) **continuous at 0**. Then for every metric d, $\alpha \circ d$ is a **topologically equivalent** metric.

If moreover (ii)′ α is strictly increasing on some $[0, \eta]$, then d-Cauchy $\iff$ $(\alpha \circ d)$-Cauchy.

Proof.

Same convergent sequences. If $d(x_n, a) \to 0$, then $\alpha(d(x_n, a)) \to 0$ by continuity at 0. If $x_n \not\to a$ in d, there are $\eta > 0$ and a subsequence with $d(x_{n_i}, a) \geq \eta$. Then $\alpha(d(x_{n_i}, a)) \geq \alpha(\eta) > 0$ (increasing and definite), so $x_n \not\to a$ in $\alpha \circ d$. Apply Equivalence via convergence.

Cauchy part. (⇒): given ε, continuity at 0 gives η with $s < \eta \Rightarrow \alpha(s) < \varepsilon$. Apply the d-Cauchy condition with η. (⇐): given $\varepsilon > 0$, $\alpha(\varepsilon) > 0$. Choose N with $\alpha(d(x_n, x_m)) < \alpha(\varepsilon)$ for $n, m \geq N$. If $d(x_n, x_m) \geq \varepsilon$, then $\alpha(d(x_n, x_m)) \geq \alpha(\varepsilon)$ (increasing), a contradiction. So $d(x_n, x_m) < \varepsilon$. ∎

Remarks: these conditions make α continuous **everywhere**, since $0 \leq \alpha(s+t) - \alpha(s) \leq \alpha(t)$. Conditions (i), (iii), (iv) are also **necessary**; perhaps surprisingly, (ii) is not.

Standard examples: $\alpha_0(s) = cs$, $\alpha_1(s) = \min\{s, 1\}$, $\alpha_2(s) = \frac{s}{1+s}$, $\alpha_3(s) = \tan^{-1} s$, $\alpha_4(s) = s^{1/p}$.

Corollary: every metric d has a topologically equivalent **bounded** metric with the same Cauchy sequences (use α_1, α_2 or α_3).

Products and metric subspaces revisited

PROPOSITION (CONVERGENCE IN COUNTABLE PRODUCT SPACES)

Let d_i' be topologically equivalent to d_i with $\sum_i d_i'\text{-}\operatorname{diam} S_i < \infty$, and let $d(x, y) = \sum_i d_i'(x_i, y_i)$ on $S = \prod S_i$.

(a) $x^{(n)} \to a \iff x_i^{(n)} \to a_i$ for every i (coordinatewise convergence).

(b) If, in addition, d_i' and d_i have the same Cauchy sequences, then d-Cauchy $\iff$ coordinatewise d_i-Cauchy.

Proof. (⇐) Given ε, choose K with the **tail** $\sum_{k>K} d_k'\text{-}\operatorname{diam} < \varepsilon$. Then choose N so that the first K coordinates are each within ε/K. This gives $d(x^{(n)}, a) < 2\varepsilon$. (⇒) $d_i'(x_i^{(n)}, a_i) \leq d(x^{(n)}, a)$. ∎

Finite products with the sum metric: convergence and Cauchyness are both coordinatewise.

BALLS IN A METRIC SUBSPACE

In $\tilde{A} = (A, d|_{A\times A})$: $\tilde{B}_r(a) = A \cap B_r(a)$ and $\tilde{B}_r[a] = A \cap B_r[a]$.

OPENNESS/CLOSEDNESS IS ALWAYS "IN" A SPACE

$A = [0, 1[$ is both open and closed in $\tilde{A}$ (it is the whole space), but neither open nor closed in $\mathbb{R}$.

In $\tilde{\mathbb{Z}} \subset \mathbb{R}$, every subset is both open and closed, because $\tilde{B}_1(k) = \{k\}$ makes singletons open.

THEOREM (CLOSED AND OPEN SETS OF A METRIC SUBSPACE)

For $X \subset A$: X is closed in $\tilde{A} \iff X = A \cap F$ for some F closed in E. Likewise, X is open in $\tilde{A} \iff X = A \cap U$ for some U open in E (Exercise 4.10).

Proof. (Closed, ⇒) Take $F = \overline{X}$ (closure in E). Then $X \subset A \cap F$. If $p \in A \cap F$, some $x_n \in X$ has $x_n \to p$. Since $p \in A$ and X is closed in $\tilde{A}$, $p \in X$. (⇐) If $x_n \in X$ and $x_n \to a \in A$, then (x_n) lies in F, which is closed in E, so $a \in F$. Hence $a \in A \cap F = X$. ∎

Exercises for Section 4, with full solutions

EXERCISE 4.1 (OPEN SUBSETS OF DISCRETE SPACES)

5 marks

▾ Solution

Every subset is open. For $p \in U$, $B_1(p) = \{x : d(x, p) < 1\} = \{p\} \subset U$. (Any radius $r \leq 1$ works.) ∎

EXERCISE 4.2 (OPEN MINUS FINITE IS OPEN)

10 marks

▾ Solution

Let $p \in U = V \setminus F$. Since V is open, $B_{r_0}(p) \subset V$ for some $r_0 > 0$. Let $r = \min(\{r_0\} \cup \{d(p, q) : q \in F\})$. This is a minimum of **finitely many positive** numbers ($p \notin F$, so $d(p, q) > 0$), hence $r > 0$. If $x \in B_r(p)$, then $x \in V$, and $x \neq q$ for every $q \in F$ (as $d(x, p) < r \leq d(q, p)$). So $B_r(p) \subset U$. ∎

Alternative: finite sets are closed (finite unions of closed singletons), so $V \setminus F = V \cap F^{\complement}$ is an intersection of two open sets. "Finite" is essential: $\mathbb{R} \setminus \mathbb{Q}$ is not open.

EXERCISE 4.3 (INVERSE IMAGES OF DIRECT IMAGES AND VICE VERSA)

10 marks

▾ Solution

(a) $f^{-1}(f(A)) = \{s : \exists a \in A,\ f(s) = f(a)\} \supset A$ (take $s = a$). $f(f^{-1}(B)) = B \cap \operatorname{Ran} f \subset B$.

(b) $f(x) = x^2$ on $\mathbb{R}$. With $A = [0,1]$: $f^{-1}(f(A)) = [-1,1] \supsetneq A$. With $B = [-1,1]$: $f(f^{-1}(B)) = [0,1] \subsetneq B$.

(c) If f is injective, $f(s) = f(a)$ forces $s = a$, so $f^{-1}(f(A)) = A$. Conversely, if equality holds for all A, take $A = \{s_1\}$: if $f(s_2) = f(s_1)$ then $s_2 \in f^{-1}(f(\{s_1\})) = \{s_1\}$. So f is injective.

(d) If f is surjective, $B \cap \operatorname{Ran} f = B$. Conversely, take $B = T$: $T = f(f^{-1}(T)) = f(S) = \operatorname{Ran} f$. ∎

EXERCISE 4.4 (CLOSED SUBSETS OF DISCRETE SPACES)

5 marks

▾ Solution

Every subset is closed: its complement is open by Exercise 4.1. So in a discrete space every set is both open and closed. ∎

EXERCISE 4.5 (GLOBAL CONTINUITY VIA CLOSED SETS)

10 marks

▾ Solution

Use $f^{-1}(E' \setminus C) = E \setminus f^{-1}(C)$ (Exercise 1.9) and "closed $\iff$ complement open".

(i)⇒(ii): if C is closed, $E' \setminus C$ is open, so $f^{-1}(E' \setminus C) = E \setminus f^{-1}(C)$ is open (global continuity). Hence $f^{-1}(C)$ is closed.

(ii)⇒(i): if V is open, $E' \setminus V$ is closed, so $E \setminus f^{-1}(V)$ is closed and $f^{-1}(V)$ is open. Global continuity via open sets gives continuity. ∎

EXERCISE 4.6 (EQUIVALENCES OF METRICS ARE EQUIVALENCE RELATIONS)

10 marks

▾ Solution

Reflexive: $\mathrm{id} : (S, d) \to (S, d)$ is an isometry, hence bi-Lipschitz (constant 1) and a homeomorphism.

Symmetric: the inverse of $\mathrm{id} : (S, d_1) \to (S, d_2)$ is $\mathrm{id} : (S, d_2) \to (S, d_1)$. Both definitions require the map and its inverse to be continuous (resp. Lipschitz), which is symmetric in d_1, d_2.

Transitive: if $d_1 \sim d_2$ and $d_2 \sim d_3$, then $\mathrm{id}_{13} = \mathrm{id}_{23} \circ \mathrm{id}_{12}$. Compositions of continuous maps are continuous: $x_n \to p \Rightarrow \mathrm{id}(x_n) \to p \Rightarrow \ldots$, or via preimages of open sets $(g \circ f)^{-1}(V) = f^{-1}(g^{-1}(V))$. Compositions of Lipschitz maps are Lipschitz: $d_3 \leq L_2 d_2 \leq L_2 L_1 d_1$. Apply the same to the inverses. ∎

EXERCISE 4.7 (EQUIVALENCE OF METRICS AND COMPARISON OF BALLS)

▾ Solution

(a) (i)⇒(ii): $\mathrm{id} : (S, d_1) \to (S, d_2)$ is continuous at p, so for $R > 0$ some r_1 has $B^1_{r_1}(p) \subset B^2_R(p)$. Symmetrically, some r_2 has $B^2_{r_2}(p) \subset B^1_R(p)$. Take $r = \min\{r_1, r_2\}$.

(ii)⇒(iii): let A be d_1-open and $p \in A$. Then $B^1_R(p) \subset A$ for some R, and (ii) gives r with $B^2_r(p) \subset B^1_R(p) \subset A$. So A is d_2-open, and symmetrically the other way.

(iii)⇒(i): the preimage under $\mathrm{id} : (S, d_1) \to (S, d_2)$ of a d_2-open V is V, which is d_1-open by (iii). By global continuity, id is continuous. The same argument applies to its inverse.

(b) (i)⇒(ii): suppose $d_2 \leq L d_1$ and $d_1 \leq K d_2$ with $L, K > 0$. If $d_1(x, p) < R/L$ then $d_2(x, p) < R$, so $B^1_{mR}(p) \subset B^2_R(p)$ with $m = 1/L$. If $d_2(x, p) < R$ then $d_1(x, p) < KR$, so $B^2_R(p) \subset B^1_{MR}(p)$ with $M = K$.

(ii)⇒(i): fix x, p. For every $R > d_1(x, p)/m$ we have $x \in B^1_{mR}(p) \subset B^2_R(p)$, so $d_2(x, p) < R$. Letting $R \downarrow d_1(x, p)/m$ gives $d_2 \leq \frac{1}{m} d_1$. Similarly, for every $R > d_2(x, p)$, $x \in B^1_{MR}(p)$, so $d_1(x, p) < MR$, giving $d_1 \leq M d_2$. Hence id is bi-Lipschitz. ∎

EXERCISE 4.8 (D_{TGV} IS NOT EQUIVALENT TO THE STANDARD METRIC ON $\mathbb{R}$)

10 marks

▾ Solution

(a) For $x \neq y$: $|x-y| \leq |x| + |y| = d_{TGV}(x,y)$. So $g = \mathrm{id} : E_2 \to E_1$ is Lipschitz with constant ≤ 1.

(b) For $x \neq 0$, $d_{TGV}(x,0) = |x| + 0 = |x - 0|$. Given ε, take δ=ε: $|x| < \delta \Rightarrow d_{TGV}(x,0) < \varepsilon$. So f is continuous at 0.

(c) For $a \neq 0$, let $x_n = a + \frac{1}{n} \to a$ in E_1. Then $d_{TGV}(x_n, a) = |x_n| + |a| \to 2|a| \neq 0$, so $f(x_n) \not\to f(a)$, and f is not (sequentially) continuous at a. (Equivalently, $B^{TGV}_{|a|}(a) = \{a\}$, so $\{a\}$ is open in E_2 but not in E_1.)

Deduce: $\mathrm{id} : E_1 \to E_2$ is not continuous, so the metrics are **not topologically equivalent**, and hence not Lipschitz equivalent either. ∎

EXERCISE 4.9 (CONTINUITY OF THE METRIC)

▾ Solution

Apply the Rearranged Triangle Inequality twice: $|d(x,y) - d(u,v)| \leq |d(x,y) - d(u,y)| + |d(u,y) - d(u,v)| \leq d(x,u) + d(y,v) = d'((x,y),(u,v))$. So d is **Lipschitz with constant ≤ 1**, and Lipschitz ⇒ continuous. ∎

EXERCISE 4.10 (OPEN SETS OF A METRIC SUBSPACE)

▾ Solution

(ii)⇒(i): let $X = A \cap U$ with U open in E, and $p \in X$. Then $B_r(p) \subset U$ for some r, so $\tilde{B}_r(p) = A \cap B_r(p) \subset A \cap U = X$.

(i)⇒(ii): for each $p \in X$ choose $r_p > 0$ with $\tilde{B}_{r_p}(p) \subset X$, and let $U = \bigcup_{p \in X} B_{r_p}(p)$. U is open in E (a union of open balls). Then $A \cap U = \bigcup_p (A \cap B_{r_p}(p)) = \bigcup_p \tilde{B}_{r_p}(p)$. This contains every $p \in X$ and is contained in X, so it equals X. ∎

Alternatively, use complements together with the closed-set theorem: $A \setminus X$ is closed in $\tilde{A}$, so $A \setminus X = A \cap F$; then $X = A \cap F^{\complement}$.

EXERCISE 4.11 (SEPARABILITY OF $C(I;\mathbb{R})$, AGAIN)

10 marks

$A = \bigcup_{n\geq 0} A_n$, where A_n is the set of continuous g, linear on each $[k/2^n, (k+1)/2^n]$, with $g(j/2^n) \in \mathbb{Q}$.

▾ Solution

(a) The map $A_n \to \mathbb{Q}^{1+2^n}$, $g \mapsto (g(0), g(1/2^n), \dots, g(1))$, is **injective**: a function that is linear on each piece is determined by its values at the nodes (linear interpolation). $\mathbb{Q}^{1+2^n}$ is countable, so A_n is countable. A countable union of countable sets is countable, so A is countable.

(b) Let $f \in C(I;\mathbb{R})$ and $\varepsilon > 0$. By the **Uniform Continuity Theorem** there is δ with $|x-y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$. Choose n with $2^{-n} < \delta$, and rationals q_j with $|q_j - f(j/2^n)| < \varepsilon$ (density of $\mathbb{Q}$). Let $g \in A_n$ interpolate the q_j.

For $x \in [x_k, x_{k+1}]$ (with $x_k = k/2^n$), write $x = (1-\lambda)x_k + \lambda x_{k+1}$ with $\lambda \in [0,1]$. Then

$$|g(x) - f(x)| = |(1-\lambda)(q_k - f(x)) + \lambda(q_{k+1} - f(x))| \leq (1-\lambda)|q_k - f(x)| + \lambda|q_{k+1} - f(x)|.$$

Each term satisfies $|q_k - f(x)| \leq |q_k - f(x_k)| + |f(x_k) - f(x)| < 2\varepsilon$ (since $|x - x_k| \leq 2^{-n} < \delta$). So $|g - f| < 2\varepsilon$ everywhere, and $d_{\sup}(f,g) \leq 2\varepsilon$. Hence every ball meets A, and A is dense.

A countable dense subset exists, so E is **separable**. ∎

EXERCISE 4.12 (UNCOUNTABILITY OF $\mathbb{R}$, AGAIN)

10 marks

$\psi : \mathbb{R} \to \{0,1\}^{\mathbb{N}}$, $x \mapsto (F(\lfloor 10x \rfloor), F(\lfloor 10^2 x \rfloor), \dots)$, where $F(m) = 0$ for m even and 1 for m odd. (a) Show ψ is surjective. (b) Deduce there is no surjection $\mathbb{N} \to \mathbb{R}$.

▾ Solution

Here $\lfloor p \rfloor = \max\{m \in \mathbb{Z} : m \leq p\}$, the integer part.

(a) Given $a = (a_k) \in \{0,1\}^{\mathbb{N}}$, let $x = \sum_{k\geq 1} a_k 10^{-k}$. Then

$$10^n x = \underbrace{\sum_{k\leq n} a_k 10^{n-k}}_{=:m_n \in \mathbb{Z}} + \underbrace{\sum_{k>n} a_k 10^{n-k}}_{=:r_n \in [0,1/9]}.$$

Since $0 \leq r_n \leq \sum_{j\geq 1} 10^{-j} = \frac{1}{9} < 1$, we get $\lfloor 10^n x \rfloor = m_n$. Also $m_n = a_n + 10(\dots)$, so $m_n \equiv a_n \pmod 2$ and $F(m_n) = a_n$. Hence $\psi(x) = a$, and ψ is surjective.

(b) If $\sigma : \mathbb{N} \to \mathbb{R}$ were surjective, then $\psi \circ \sigma : \mathbb{N} \to \{0,1\}^{\mathbb{N}} \cong \mathcal{F}(\mathbb{N}; \{0,1\})$ would be a surjection, contradicting **Cantor's Theorem** (no surjection $S \to \mathcal{F}(S;\{0,1\})$). Since $\mathbb{R} \neq \emptyset$, this means $\mathbb{R}$ is not countable. ∎

IN-TEXT: WHY EVERY SUBSET OF $\tilde{\mathbb{Z}}$ IS OPEN (AND CLOSED)

▾ Answer

Every subset $X \subset \mathbb{Z}$ is a union of singletons, $X = \bigcup_{k \in X} \{k\}$. Each singleton is the open ball $\tilde{B}_1(k)$, and unions of open sets are open. So X is open; and since $\mathbb{Z} \setminus X$ is open too, X is also closed. This explains the "It suffices to show… (Why?)" step in the notes.

Trap questions: true or false?

Examiners like these. Decide first, then open each one to check. Every "false" comes with a counterexample, which is what an answer needs.

▼ 1. Every subset of a metric space is either open or closed.

False. $[0,1[$ and $\mathbb{Q}$ in $\mathbb{R}$ are neither. Some sets are both: $\emptyset$, E, and every subset of a discrete space.

▼ 2. An arbitrary intersection of open sets is open.

False. $\bigcap_n]-1/n, 1/n[= \{0\}$. Only *finite* intersections of open sets are open; *arbitrary unions* are open.

▼ 3. An arbitrary union of closed sets is closed.

False. $\bigcup_n [1/n, 1] =]0,1]$. Only *finite* unions of closed sets are closed; *arbitrary intersections* are closed.

▼ 4. The closure of the open ball $B_r(a)$ is always the closed ball $B_r[a]$.

False. In a discrete space, $\overline{B_1(a)} = \{a\}$ but $B_1[a] = E$. Only $\overline{B_r(a)} \subset B_r[a]$ holds in general. Equality holds in normed spaces.

▼ 5. $\operatorname{diam} B_r(a) = 2r$ in every metric space.

False. Only $\leq 2r$. In a discrete space $\operatorname{diam} B_{1/2}(a) = 0$.

▼ 6. $f(A \cap B) = f(A) \cap f(B)$ for every function.

False. Only $\subset$ holds. Counterexample: a constant function with $A = \{0\}$, $B = \{1\}$. Equality holds when f is injective. Inverse images always preserve $\cap$, $\cup$ and complements.

▼ 7. Writing $f^{-1}(C)$ requires f to be bijective.

False. The inverse image $\{s : f(s) \in C\}$ is defined for every function.

▼ 8. A symmetric, transitive relation is reflexive.

False. The empty relation on a nonempty set, or $\{(1,1)\}$ on $\{1,2\}$. The "proof" assumes each x is related to something.

▼ 9. A continuous bijection between metric spaces is a homeomorphism.

False. $\mathrm{id} : (\mathbb{R}, d_{\text{disc}}) \to (\mathbb{R}, |\cdot|)$ is a continuous bijection, but its inverse is not continuous.

▼ 10. Continuous on $]0,1]$ implies uniformly continuous.

False. $1/x$ is a counterexample. It is true on **closed and bounded** intervals (Uniform Continuity Theorem).

▼ 11. Topologically equivalent metrics have the same Cauchy sequences.

False. $|\tan^{-1} y - \tan^{-1} x|$ vs $|y-x|$ on $\mathbb{R}$: $x_n = n$ is Cauchy for the first only. Lipschitz equivalent metrics do have the same Cauchy sequences.

▼ 12. Topologically equivalent metrics have the same bounded sets.

False. d vs $\min\{d, 1\}$ on $\mathbb{R}$.

▼ 13. Every Cauchy sequence in a metric space converges.

False in general. In $(]0,1], |\cdot|)$, the sequence $1/n$ is Cauchy but has no limit *in the space*. It is true in $\mathbb{R}$ (completeness of $\mathbb{R}$).

▼ 14. Every metric on a vector space comes from a norm.

False. The discrete metric and the $\tan^{-1}$ metric are bounded, but a norm-metric on a nonzero space is unbounded.

▼ 15. If D and A are dense in E, then $D \cap A$ is dense.

False. $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$. It is true if one of them is **open** and dense.

▼ 16. A metric subspace inherits the "shape" of the space, like linear subspaces do.

False. $\{e_1, \ldots, e_n\} \subset \mathbb{R}^n$ is the discrete metric times $\sqrt{2}$.

▼ 17. "$[0,1[$ is open" is a meaningful statement on its own.

Not without saying "in which space". It is open (and closed) in itself, and neither in $\mathbb{R}$.

▼ 18. A countable union of countable sets is countable.

True, using countable choice to pick the surjections ψ_n.

▼ 19. $\mathcal{P}(\mathbb{N})$ is countable.

False (Cantor). But $\mathcal{P}_{00}(\mathbb{N})$, the set of finite subsets, **is** countable.

▼ 20. The LUB Principle is a theorem proved from the field axioms.

False. It is an **axiom** (completeness axiom). The Archimedean property is *derived* from it. ($\mathbb{Q}$ is an ordered field without LUB.)

▼ 21. Lipschitz ⇒ uniformly continuous ⇒ continuous, and both converses fail.

True. $\sqrt{x}$ on $[0,1]$ is uniformly continuous (UCT) but not Lipschitz. $1/x$ on $]0,1]$ is continuous but not uniformly continuous.

▼ 22. If $x_n < y_n$ for all n and both converge, then $\lim x_n < \lim y_n$.

False. Only ≤ survives: $0 < 1/n$, but both limits are 0.

▼ 23. An isometry need not be injective.

False. Isometries are always injective, since $d(x,y) = d(f(x),f(y)) = 0 \Rightarrow x = y$. They need not be surjective.

▼ 24. Continuous maps send open sets to open sets.

False. A constant map. It is *preimages* of open sets that are open.

▼ 25. $\operatorname{Int}\mathbb{Q} = \emptyset$, $\overline{\mathbb{Q}} = \mathbb{R}$ and $\partial\mathbb{Q} = \mathbb{R}$ in $\mathbb{R}$.

True. This uses density of both $\mathbb{R} \setminus \mathbb{Q}$ and $\mathbb{Q}$.

Flashcards: definitions and statements to know word for word

Say the answer out loud before you open the card. If you hesitate, review that card again.

▼ Metric (all axioms, named)

$d : S \times S \to [0, \infty[$ with (MS1) $d(x,y) = 0 \iff x = y$; (MS2) Symmetry; (MS3) Triangle Inequality $d(x,z) \leq d(x,y) + d(y,z)$.

▼ Norm (N1)–(N3)

Definiteness $\|v\| = 0 \Rightarrow v = 0$; Homogeneity $\|\lambda v\| = |\lambda| \|v\|$; Triangle inequality.

▼ Injective / surjective

$f(s_1) = f(s_2) \Rightarrow s_1 = s_2$ / $\forall t \, \exists s : \; f(s) = t$ ($\operatorname{Ran} f = T$).

▼ Left / right inverse $\longleftrightarrow$?

Left inverse $\iff$ injective (if $S \neq \emptyset$ or $T = \emptyset$). Right inverse $\iff$ surjective (needs **AC**).

▼ Axiom of Choice (two forms)

A product of nonempty sets is nonempty. Equivalently, every family of nonempty subsets has a choice function.

▼ Equivalence relation; partition

Reflexive, symmetric, transitive. Partition: covers S, blocks nonempty, distinct blocks disjoint. The two notions correspond bijectively.

▼ Canonical decomposition

$f = i \circ \tilde{f} \circ q$: quotient (surjection), then $\tilde{f} : [s] \mapsto f(s)$ (bijection onto $\operatorname{Ran} f$), then inclusion (injection).

▼ Countable

There is an injection $S \to \mathbb{N}$. For $S \neq \emptyset$, equivalently a surjection $\mathbb{N} \to S$. Equivalently, finite or bijective to $\mathbb{N}$.

▼ Cantor's Theorem

No surjection $S \to \mathcal{P}(S)$. Proof via $D = \{s : s \notin \phi(s)\}$. Also no surjection $S \to \mathcal{F}(S; \{0,1\})$.

▼ Schröder–Bernstein

Injections $S \to T$ and $T \to S$ ⇛ a bijection $S \to T$.

▼ LUB Principle

A nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound. It is an **axiom**.

▼ Approximation of suprema

$\forall \varepsilon > 0\ \exists a \in A :\ \sup A - \varepsilon < a \leq \sup A$. Sequence version: some increasing $(a_n) \subset A$ has $a_n \to \sup A$.

▼ Archimedean property (4 forms)

$\mathbb{Q}$ is order dense; $\forall x > 0\ \exists n :\ 1/n < x$; $\forall x > 0, y\ \exists n :\ nx \geq y$; $\mathbb{N}$ is unbounded.

▼ Bolzano–Weierstrass

Every sequence in a closed bounded interval J has a subsequence converging to a point of J.

▼ Completeness of $\mathbb{R}$

Convergent $\iff$ Cauchy $\iff$ bounded with $\liminf = \limsup$.

▼ Uniform Continuity Theorem

Continuous on a closed bounded interval ⇒ uniformly continuous, bounded, attains max and min.

▼ Lipschitz map; Lip f

$d(f(x), f(y)) \le M\, d(x, y)$; $\operatorname{Lip} f = \sup_{x \neq y} \frac{d(f(x), f(y))}{d(x, y)}$. Strict contraction: $\operatorname{Lip} f < 1$.

▼ Isometry; isometric spaces

Preserves distances exactly. Spaces are isometric if there is an isometric **bijection**.

▼ Diameter; bounded set

$\operatorname{diam} A = \sup d(x, y)$; bounded iff it is finite.

▼ Open ball / closed ball / sphere

$d(x, a) < r$ / $\le r$ / $= r$.

▼ Open set

$\forall p \in U\ \exists r > 0 :\ B_r(p) \subset U$.

▼ Point of closure (3 equivalent forms)

Every ball meets A; a sequence in A converges to p; $\operatorname{dist}_A(p) = 0$.

▼ Closed set (2 forms)

$\overline{A} = A$. Equivalently, limits of convergent sequences in A lie in A. Equivalently, the complement is open.

▼ Interior; closure as extremal sets

$\operatorname{Int} A$ is the largest open subset of A. $\overline{A}$ is the smallest closed superset.

▼ Boundary

$\partial A = \overline{A} \setminus \operatorname{Int} A = \overline{A} \cap \overline{A^{\complement}}$. It is closed.

▼ Dense; separable

$\overline{D} \supset A$. Separable: has a countable dense subset.

▼ Global continuity

f continuous $\iff$ preimage of every open set is open $\iff$ preimage of every closed set is closed $\iff$ sequentially continuous.

▼ Homeomorphism (H1–H3)

Continuous, bijective, continuous inverse.

▼ Topological vs Lipschitz equivalence

id_S is a homeomorphism vs bi-Lipschitz ($d_2 \le L d_1$, $d_1 \le K d_2$). The second implies the first, not conversely.

▼ When is α∘d an equivalent metric?

α definite, increasing, subadditive, continuous at 0. Examples: cs, $\min\{s, 1\}$, $s/(1+s)$, $\tan^{-1} s$, $s^{1/p}$.

▼ Open/closed sets of a subspace

X open (closed) in $\tilde{A} \iff X = A \cap U$ with U open (closed) in E.

▼ Open subsets of $\mathbb{R}$

Every open subset of $\mathbb{R}$ is a countable union of open intervals with rational endpoints.

Glossary & notation

Symbol / term	Meaning
$\mathbb{N}$, $\mathbb{Z}_+$	$\{1,2,\dots\}$, $\{0,1,2,\dots\}$
$]a,b[$, $[a,b[$	open interval, half-open interval
$\mathcal{F}(S;T)$, T^S	all functions $S \to T$; identified with the product $\prod_{s\in S} T$
$\mathcal{P}(S)$, $\mathcal{P}_{00}(S)$	power set; finite subsets
$A \Delta B$	symmetric difference
$\operatorname{Ran} f$, $f(A)$, $f^{-1}(C)$	range, direct image, inverse image
$f\|_A$, $f\|^B$	restriction, co-restriction
$\mathbf{1}_A$, $\iota_{A\to S}$, id_S	indicator, inclusion, identity
$S/\!\sim$, $[a]_\sim$, q	quotient set, equivalence class, quotient map
U_A	set of upper bounds of A
$C(A;\mathbb{R})$, C_{Lip}, $\mathcal{F}_b$, $\mathcal{R}(J)$, C^1	continuous, Lipschitz, bounded, Riemann integrable, continuously differentiable functions
$L(U;V)$, V^{dual}	linear maps; dual space
$\ell^p(N)$, $\ell^\infty(S;E)$	$(\mathbb{R}^N, d_p)$; $(\mathcal{F}_b(S;E), d_{\sup})$
$B_r(a)$, $B_r[a]$, $S_r(a)$	open ball, closed ball, sphere
diam, dist_A, Lip	diameter, distance from A, Lipschitz constant
$\mathcal{O}(E)$, $\mathcal{C}(E)$	open sets, closed sets
$\operatorname{Int} A$, $\overline{A}$, ∂A, $A^{\complement}$	interior, closure, boundary, complement
$\tilde{A}$, $\tilde{B}_r(a)$	metric subspace and its balls
TFAE, WLOG, ∎	the following are equivalent; without loss of generality; end of proof

Built from the MATH6310 lecture notes, Sections 0–4 (edition 07.ix.2026). Where these notes add an explanation, example or solution, it is written to be consistent with the notes' definitions and conventions.